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PROJECT TITLE

AGENTIC AI FOR ADAPTIVE CURRICULUM FORMATION

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ABSTRACT

This study proposes an Agentic Artificial Intelligence (AI) system to overcome the scalability constraints of human advising and the static nature of current recommendation systems. Unlike passive systems, this Agentic design enables academic experts to change criteria and weights dynamically without any reprogramming in the backend. The system uses an advanced "Reasoning Core" supported by four parallel hybrid MCDM methods, namely q-ROF-AHP, BWM-VIKOR, SWARA-MOORA-3NAG, and LTSF-CRITIC-EDAS to handle multi-dimensional data about students, including SPM results, university grades, technical skills, and RIASEC personality types. The reliability test performed using IIUM's KICT Computer Science students showed excellent reliability, achieving excellent consistency with experts' opinions and successfully uncovering hidden aptitudes in students that might not be recognized using conventional methods. This work, therefore, fills the gap between static educational software and dynamic decision support, providing a scalable "Universal Academic Advisor."

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LIST OF ABBREVIATIONS

3NAG	Three Normalisation Aggregation
ADE	Application Development Engineering
AHP	Analytic Hierarchy Process
AI	Artificial Intelligence
AS	Appraisal Score
AV	Average Solution
BWM	Best-Worst Method
CRITIC	Criteria Importance Through Intercriteria Correlation
DE	Data Engineering
EDAS	Evaluation based on Distance from Average Solution
KICT	Kulliyyah of Information and Communication Technology
LT-SF/LTSF	Linguistic T-Spherical Fuzzy
MCDM	Multi-Criteria Decision-Making
MOORA	Multi-Objective Optimisation on the basis of Ratio Analysis
NDA	Negative Distance from Average
NDC	Networking and Data Communication
OAG	Overall Absolute Ordering
PDA	Positive Distance from Average
q-ROF	q-Rung Orthopair Fuzzy
q-ROFN	q-Rung Orthopair Fuzzy Number
RIASEC	Realistic, Investigative, Artistic, Social, Enterprising, and Conventional
SDS	Security in Digital System
SPM	Sijil Pelajaran Malaysia (Malaysian Certificate of Education)
SWARA	Step-wise Weight Assessment Ratio Analysis
VIKOR	ViseKriterijumska Optimizacija I Kompromisno Resenje

CHAPTER ONE

INTRODUCTION

1.1 Background of Study

In the rapidly evolving landscape of higher education, choosing the most suitable specialized major is a very critical moment that will determine the future of a student's working career. Many students struggle during their study or even upon entering their working era when they choose the wrong path, often leading to "interest confusion" where their academic choices do not align with their actual career aspirations. This complex decision is not easy to be made by the students themselves and they usually need experts' suggestions and guidance to help them to choose the right path. By using the traditional method, students are required to seek personalized guidance through manual and often time-consuming processes. They typically must schedule appointments with faculty experts such as Academic Advisor or even visit the university counselling centres to discuss their options. All these consultations are inherently limited by the availability of human experts, which may result in delays or limited access. Students might face the risk of receiving inconsistent guidance if multiple experts are involved, leading to decisions made based on incomplete information.

Additionally, undergraduate students also face another challenge in making the right choice in choosing their specialisation/major, it is when they lack awareness regarding the specific metrics they should evaluate. Even though they know that their grades are very important, they rarely possess the expert knowledge to define which specific criteria are important to which specialisations/majors. Some aspects might be important for the Data Engineering field but those same aspects might not be so useful in the Networking field. They also often struggle to answer critical questions: Does the high school foundations still matter? Is introverted personality a barrier to becoming a network engineer? Are technical skills more critical than Cumulative Grade Point Average (CGPA)? Without a proper system, students frequently overlook these vital indicators, making high-stakes decisions based on peer pressure or passing trends rather than data-driven self-assessment.

To address these issues, the current research proposes the utilization of Agentic Artificial Intelligence (AI) as the underlying architecture of the latest generation of educational suggestion systems. Agentic AI is radically different from the traditional system

as they don't only process information but instead try to model the autonomy and intelligence of a human expert. At the core of the intelligence of this agent is the utilization of Multi-Criteria Decision Making (MCDM). In practical academic advising, the decision to be made involves the engagement of complex trade-offs. For example, a student might have technical skills that match a certain career path but have a certain personality that contradicts the requirements of the said profession. In order to mathematically represent this human-like subtlety, the system uses sophisticated hybrid MCDM models, namely q-Rung Orthopair Fuzzy Analytic Hierarchy Process (q-ROF-AHP), Best-Worst Method + ViseKriterijumska Optimizacija I Kompromisno Resenje (BWM+VIKOR), Step-wise Weight Assessment Ratio Analysis + Multi-Objective Optimisation - 3 Non-Additive Groups (SWARA+MOORA-3NAG), and Linguistic T-Spherical Fuzzy – Criteria Importance Through Intercriteria Correlation – Evaluation based on Distance from Average Solution (LTSF-CRITIC-EDAS). These mathematical computations enable the agent to assign weights to the competing criteria and to prioritize alternatives with a complexity that cannot be replicated by a rule-based system. By doing so, the system is able to assess four key criteria that are the pillars of the decision-making process, which are the SPM outcome, the results of the last semester (University subjects outcome), technical skills, and personality traits (RIASEC personality traits). This Agentic approach enables every student to get an expert-level assessment without requiring an actual appointment, thus providing access to quality academic advising.

1.2 Problem Statement

This research aims to address three fundamental problems that preventing university students from receiving the best specialisation recommendations within the currently existing system:

1. **Inflexibility of Existing Recommendation Systems:** Current curriculum recommendation systems rely on hard-coded logic that prevents adaptation by experts. Changes to curriculum rules or faculty requirements necessitate manual backend reprogramming by developers, which results in the system providing outdated recommendations that no longer follow current faculty policies (Kodors et

al., 2025).

2. **Limitations of Human Academic Advising:** human academic advisors lack the cognitive capacity to simultaneously process and weight high-dimensional datasets (SPM results, University grades, Technical Skills, and RIASEC traits) for every student. This results in heuristic-based or generic advice rather than data-driven optimisation, often leading to inconsistent student guidance (Tyton Partners, 2022).
3. **Mismatch Between Student Interests and Abilities:** Students frequently suffer from conflicting interests, selecting specialisations based on peer pressure or perceived trends rather than empirical evidence of their technical strengths. Existing systems fail to objectively audit this disconnect, leading to poor academic fit and increased dropout risks (Heß, 2023).

1.3 Research Objectives

This research is undertaken with the following specific objectives to guide the development of the agentic AI curriculum recommendation system:

1. To develop an agentic AI recommendation system based on the four MCDM methods (q-ROF-AHP, BWM+VIKOR, SWARA+MOORA-3NAG, and LTSF-CRITIC-EDAS) for students.
2. To evaluate the performance of the system using the four hybrid MCDM methods in simulating academic advising decisions.
3. To validate the system through comparison with expert academic advisor's judgments.

1.4 Research Questions

1. How can an AI system be architected to function as an "adaptive agent" allowing for the dynamic adjustment of curriculum rules without backend reprogramming?
2. Which hybrid MCDM algorithm provides the most robust "reasoning engine" for the agent to simulate human expert decision-making in an educational context?

3. To what extent do the recommendations of this agentic AI curriculum recommendation system align with successful student outcomes, and what does this indicate about its potential for broader application across faculties/kulliyyah?

1.5 Significance of the Study

This research bridges the gap between Educational Field and Agentic Systems.

- For the Field of AI: offers a paradigm shift from static expert systems to dynamic and user-controlled decision agents.
- For Higher Education Institutions: It provides a framework for the construction of a "Universal Academic Advisor" tool. With this tool optimized, the system can be applied to other faculties or kulliyyahs (such as Engineering and Economics) by simply feeding their respective requirements into the agent.
- For the User: It offers an immediate solution to the existing problem of assigning the appropriate Computer Science specializations to the students.

1.6 Scope of the Study

The study consists of two phases. This report will document phase one (FYP 1).

- Primary Focus (Phase 1): Development of the central part of the Agentic AI System and its validation using the KICT Computer Science Curriculum as the prototype.
- Domain Context: The prototype is restricted to the 5 specialisations available for the Bachelor of Computer Science programme in IIUM (Application Development Engineering, Artificial Intelligence, Security in Digital System, Data Engineering, Networking and Data Communication).
- Agent Capabilities: The “Agent” can be defined in terms of the capability of the system to:
 - Take dynamic inputs from an expert (Academic Advisor) on the weights of the criteria.

- It processes four dimensions of data: SPM Result, Previous Semester Result (Results of University Subjects), Technical Skills, and Personality types (RIASEC).
- Implement 4 different MCDM reasoning algorithms.
- Future Work (Phase 2): Enhancing the database structure of the system to support multi-faculty data models, thereby making the system universal.

1.7 Structure of the Report

This report is divided into four major chapters, with the addition of references and appendices, in order to systematically describe the development and validation of the Agentic AI Curriculum Recommendation System. This report outline is structured in the following manner:

- **Chapter One: Introduction**

The chapter lays the basis for the research work. It includes the background of the research, emphasizing the limitations of the conventional static systems for advising, the need for the development of "Agentic" solutions, the formulation of the problem statement, the objectives, the research questions, the importance of the research work, and the scope of the research work, specifically the Bachelor of Computer Science specialisations offered in KICT.

- **Chapter Two: Review of Previous Work**

This chapter introduces the theory and literature that are applicable to this work. It also describes the principles surrounding Agentic AI and Multi-Criteria Decision Making. It will also describe in detail the four combined MCDM methods used in the reasoning core, which include BWM+VIKOR, SWARA+MOORA+3NAG, CRITIC+EDAS, and q-ROF-AHP. Finally, this chapter will also describe the RIASEC theory and its correlation to computer science fields.

- **Chapter Three: Methodology**

This chapter describes the research design and the operational system architecture. It discusses the system flowchart, including the dual-input layer (student and expert) and the processing logic. It also describes the population and the purposive sampling

technique of 10 students for the pilot test. Finally, it describes the data collection instruments and the particular data analysis techniques utilized in validating the system, including the "Top-2 Ranking" comparative strategy.

- **Chapter Four: Results and Discussions**

This chapter will outline the empirical results of the research. It will show the system rankings for each of the 10 student profiles using all four MCDM engines. These rankings will then be compared to the "Ground Truth" rankings provided by the Academic Advisor. This chapter will conclude with a discussion on the accuracy of the system, examining "Primary" and "Secondary" rankings to assess the validity of the Agentic AI in modeling human expert opinion.

- **References and Appendices**

The report ends with a list of references used in the study and appendices that include student profile information, the project Gantt chart, and the project logbook.

CHAPTER TWO

LITERATURE REVIEW & REVIEW OF PREVIOUS WORKS

2.1 Introduction

This chapter offers a detailed overview of the previous studies that are pertinent for the development of the Agentic AI system for adaptive curriculum formation. The main emphasis is given to the MCDM methodologies and the implementation of the MCDM techniques for education and industrial applications. This review of the previous studies offers the theoretical and empirical base for the implementation of the hybrid MCDM techniques for the development of the recommendation system for the Agentic AI system. Moreover, the gap that exists in the previous studies is also identified, and the gap is the absence of adaptive and expert-modifiable AI for education that is addressed in this study.

2.2 Theoretical Framework

The theoretical framework of the present study is based on the Multi Criteria Decision Making approach and Agentic Artificial Intelligence. The former is useful in providing a framework for the evaluation of alternatives in terms of multiple, usually conflicting, criteria. In the context of academic counseling, the criteria may include academic performance, technical skills, personality, etc. The approach is useful in academic counseling, as the criteria are usually in conflict with one another (Erkan & Rouyendegh, 2014). The inclusion of Agentic Artificial Intelligence in the present study makes the expert system dynamic in the sense that the decision-making approach may be modified according to the needs of the experts without the need for programming. The present study makes use of the hybrid approach of the following four Multi Criteria Decision Making techniques:

1. Bayesian Best-Worst Method (BWM) + ViseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR)
2. Step-Wise Weight Assessment Ratio Analysis (SWARA) + Multi-Objective Optimisation by Ratio Analysis (MOORA) + Three Normalisations and Absolute Ordering (3NAG)
3. Criteria Importance Through Intercriteria Correlation (CRITIC) + Evaluation based on Distance from Average Solution(EDAS) + Linguistic T-Spherical Fuzzy (LT-SF)
4. q-Rung Orthopair Fuzzy (q-ROF) - Analytical Hierarchy Process (AHP)

The methods described above have been selected based on their proven effectiveness in dealing with complex multi-dimensional decision-making situations in various areas (Ak et al., 2021; Fernandes et al., 2023; Rasool et al., 2025; Khan et al., 2025).

2.3 Review of Related Studies

2.3.1 BWM + VIKOR

Methods Used

In the research conducted by Ak, Yucesan, and Gul in 2021, a technique referred to as BWM integrated with the VIKOR compromise ranking method was used. It is worth mentioning that the use of the Bayesian method is an extension of the traditional method of BWM, which incorporates the use of statistics to compile opinions of multiple experts in a probabilistic format. It is also worth pointing out that the method needs only $2n-3$ comparisons to be conducted per individual expert, which is a reduction in cognitive burden in relation to methods such as AHP that require pairwise comparisons to be conducted. However, the integrity of information is retained in the process. On the other hand, the use of the VIKOR method is a ranking technique that ensures that decision options are evaluated in relation to their closeness to an ideal solution, thereby ensuring that conflicts in decision-making are resolved through the use of the following indices: S (group utility representing average performance), R (individual regret indicating worst-case performance), and Q (compromise index balancing both aspects).

The methodology consists of four distinct stages. The preparation step involves the establishment of six distinct risk criteria: Occurrence, Frequency, Severity, Detectability, Cost, and PPE Sensitivity. Furthermore, five experts from the Occupational Health and Safety (OHS) domain are selected. A list of 53 distinct hazards is identified from a textile production plant. The Bayesian BWM methodology deals specifically with the weighting of the established criteria. This involves the selection of the Best and Worst criteria and the application of a standardized scale of 1–9. This information is incorporated into the Bayesian BWM algorithm. The VIKOR methodology deals specifically with the ranking of the identified hazards. This involves the assessment of the 53 identified hazards against the six established criteria. This information is incorporated into the VIKOR algorithm. The VIKOR methodology provides the indices of the identified hazards. This information is incorporated into the VIKOR algorithm. The validation of the results against the Classical BWM and

sensitivity analysis of the VIKOR indices complete the four stages of the methodology.

Flowchart

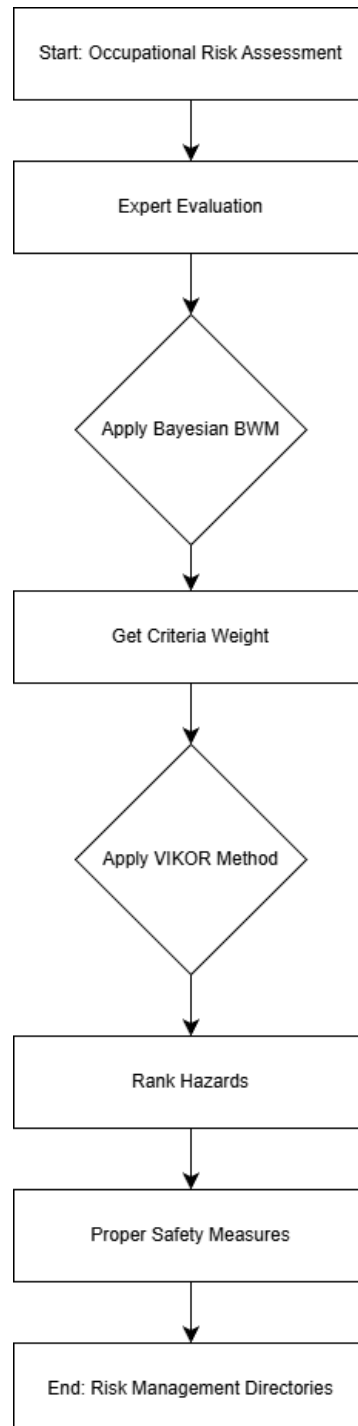


Figure 2.3.1.1: BWM–VIKOR Methodology Flowchart for Occupational Risk Assessment
(Ak et al., 2021)

The flowchart represents the visual illustration of the BWM-VIKOR methodology in its linear progression. It starts with "Occupational Risk Assessment" and then proceeds to "Expert Evaluation" to obtain the expert inputs. The flowchart then proceeds to "Apply Bayesian BWM" for the calculation of weights, then to "Apply VIKOR Method" for the ranking of the hazards. The last stages of the flowchart are "Rank Hazards" and "Propose Safety Measures," culminating in the final stage of "Risk Management Decisions."

Formula

1. Bayesian BWM Inputs:

- a. Best-to-Others Vector: $BO^k = (a_{B1}^K, a_{B2}^K, ..., a_{Bn}^K)$
- b. Others-to-Worst Vector: $OW^k = (a_{1W}^K, a_{2W}^K, ..., a_{nW}^K)$

These vectors capture each expert's (k) pairwise comparison judgments, where a_{Bj}^K represents the preference of the Best criterion over criterion j , and a_{jW}^K represents the preference of criterion j over the Worst.

2. Bayesian Hierarchical Model:

- a. Likelihood: $BO^k | w^{agg} \sim Multinomial\left(\frac{1}{\sum_j \left(\frac{w_j^{agg}}{w_j^{agg} \cdot a_{Bj}^k}\right)}, \frac{w_B^{agg}}{w_j^{agg} \cdot a_{Bj}^k}\right)$ and

$$OW^k | w^{agg} \sim Multinomial\left(\frac{1}{\sum_j \left(\frac{w_j^{agg}}{w_W^{agg} \cdot a_{jW}^k}\right)}, \frac{w_j^{agg}}{w_W^{agg} \cdot a_{jW}^k}\right)$$

- b. Prior: $w^{agg} \sim Dirichlet(a)$
- c. Posterior:

$$P(w^{agg} | \{BO^1, ..., BO^K\}, \{OW^1, ..., OW^K\}) \propto P(w^{agg}) \times \prod_{k=1}^K P(BO^k | w^{agg}) P(OW^k | w^{agg})$$

This statistical model combines all expert judgments. The Dirichlet prior represents initial beliefs about weight distributions. The multinomial likelihood functions model how observed

expert comparisons (BO^k , OW^k) are generated from the aggregated weights (w^{agg}). Bayesian inference yields the posterior distribution for w^{agg} , providing the final probabilistic weights.

3. VIKOR Calculations:

a. Best/Worst Values:

i. For benefit criteria: $f_j^* = \max_i f_{ij}$, $f_j^- = \min_i f_{ij}$

ii. For cost criteria: $f_j^* = \min_i f_{ij}$, $f_j^- = \max_i f_{ij}$

b. Group Utility (S_i): $S_i = \sum_{j=1}^n w_j \cdot \frac{f_j^* - f_{ij}}{f_j^* - f_j^-}$

c. Individual Regret (R_i): $R_i = \max_j \left[w_j \cdot \frac{f_j^* - f_{ij}}{f_j^* - f_j^-} \right]$

d. Compromise Index (Q_i): $Q_i = v \cdot \frac{S_i - S^*}{S^- - S^*} + (1 - v) \cdot \frac{R_i - R^*}{R^- - R^*}$

First, the ideal (f_j^*) and anti-ideal (f_j^-) scores are found for each criterion j . S_i is a weighted, normalized sum of distances from the ideal, representing average performance. R is the maximum normalized distance for any single criterion, representing worst-case performance. Q_i is a convex combination of normalized S and R , where parameter v (often 0.5) trades off group utility against individual regret. Alternatives are ranked by ascending Q_i .

In this research, BW+VIKOR is adapted to weigh the four academic advising criteria (SPM results, university grades, technical skills, and personality traits) and rank specialisations. Its efficient expert aggregation mechanism and its inherent design for finding compromise solutions make it particularly suitable for educational decisions involving inevitable trade-offs between conflicting criteria like technical aptitude versus personality fit.

2.3.2 SWARA + MOORA + 3NAG

Methods Used

Fernandes et al. (2023) applied a hybrid framework combining SWARA, MOORA and the 3NAG method. SWARA is employed for subjective criteria weighting, where experts

first rank criteria by importance and then assess the relative importance of each criterion compared to the one ranked above it. MOORA provides the primary ranking mechanism through a ratio-based normalisation system that accommodates both benefit and cost criteria. The 3NAG method enhances the robustness of MOORA by applying three distinct normalisation techniques which are vector, linear, and max-based, and aggregating the results into an Overall Absolute Ordering (OAG), thereby eliminating the need for a separate sensitivity analysis.

The implementation is achieved through a four-phase model, which is specifically designed for engineering supplier selection. In Phase 1, SWARA Criteria Weighting, a panel of experts ranks the importance of the seven technical criteria from highest to lowest importance. To calculate the importance of each of the criteria, experts are asked, for each of the criteria, commencing with the second-most important, how much less important it is than the immediately preceding criterion, using a "comparative importance ratio" denoted as " s_j ." The SWARA algorithm then uses the ratios of importance, which are computed using the above information, to calculate the importance weights of the criteria. In Phase 2, MOORA Initial Ranking, a pre-existing technical score matrix is used, which provides a score from 0-100 on the basis of the seven criteria for all the suppliers. The MOORA algorithm, using the importance weights, performs an initial vector normalisation, followed by the computation of an initial ranking score for all the suppliers. In Phase 3, 3NAG Robustness Enhancement, two normalisation steps, which are sum-based normalisation and max-based normalisation, are performed on the decision matrix, followed by computation of an "Absolute Ordering" value for all the suppliers for each of the normalisations. The values are summed up, giving an "OAG" value for all the suppliers, which is the final ranking of the suppliers, with Supplier C being identified as the optimum choice.

Flowchart

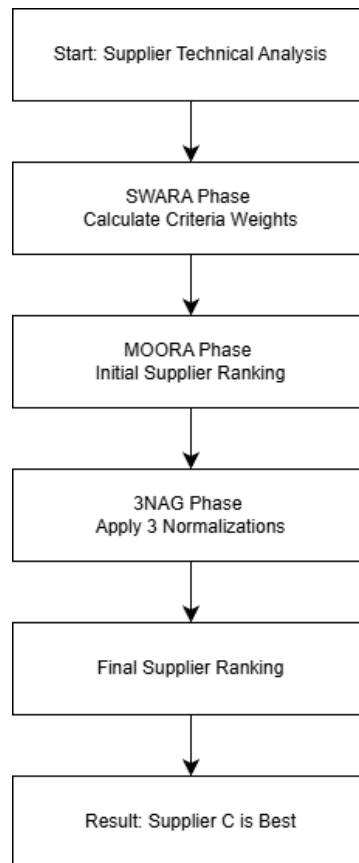


Figure 2.3.2.1: SWARA–MOORA–3NAG Methodology Flowchart for Supplier Technical Analysis (Fernandes et al., 2023)

The flowchart provided below shows a linear process from "Supplier Technical Analysis." The process begins with "SWARA Phase: Calculate Criteria Weights." This follows with "MOORA Phase: Initial Supplier Ranking." The results are then processed with "3NAG Phase: Apply 3 Normalisations," leading to "Final Supplier Ranking" and ending with "Supplier C is Best.". This visual clearly separates the weighting, initial ranking, and robustness enhancement stages.

Formula

1. SWARA

- a. Calculate Comparative Coefficient: $K_j = s_j + 1$
- b. Calculate Initial Weight: $W_j = \frac{X_j - 1}{K_j}$
- c. Calculate Final Normalized Weight: $q_j = \frac{W_j}{\sum_j W_j}$

The SWARA method calculates criterion weights through a systematic, rank-based approach where experts first order criteria by importance and then quantify the relative difference between each consecutive pair. The calculation begins with the most important criterion assigned a preliminary weight of 1. For each subsequent criterion j , a comparative coefficient, $K_j = s_j + 1$ is computed, where s_j represents the expert's judgment on how much less important criterion j is compared to the previous one in the ranking (e.g., 0.1 for 10% less important). This coefficient is then used to derive the criterion's initial weight, $W_j = \frac{X_j - 1}{K_j}$, where $X_j - 1$ is the weight of the more important preceding criterion. This recursive calculation progresses down the ranked list, systematically reducing weights based on expert-assigned importance gaps. Finally, all initial weights are normalized by dividing each by their sum, $q_j = \frac{W_j}{\sum_j W_j}$, to produce final criterion weights that sum to 1, ready for integration into the subsequent MOORA ranking process.

2. MOORA (First Normalisation & Score)

- a. First Normalisation (Vector Normalisation): $N^{Xij} = \frac{X_{ij}}{\sqrt{\sum_{j=1}^m X_{ij}^2}}$
- b. Calculate MOORA Score: $N^{Yi} = \sum_{i=1}^g N^{Xij} - \sum_{i=g+1}^n N^{Xij}$
- c. Tchebycheff Min-Max Metric: $oc = \min(i) \{ \max(i) | r_i - N^{Xi} | \}$

The MOORA method transforms raw performance scores into normalized rankings through a multi-step ratio analysis. First, vector normalisation is applied to each criterion column:

$N^{Xij} = \frac{X_{ij}}{\sqrt{\sum_{j=1}^m X_{ij}^2}}$, which converts each original score X_{ij} into a dimensionless ratio by

dividing it by the square root of the sum of squared scores for that criterion. This normalisation preserves the relative differences while eliminating scale effects across different measurement units. Next, the MOORA score for each alternative i is computed as

$$N^{Yi} = \sum_{i=1}^g N^{Xij} - \sum_{i=g+1}^n N^{Xij}, \text{ where the first sum includes all benefit criteria (to be maximized) and the second sum includes all cost criteria (to be minimized).}$$

The above score is a net score, reflecting the balance between desirable and undesirable attributes. Lastly, a Tchebycheff Min-Max metric $oc = \min(i)\{\max(i)|r_i - N^{Xi}|\}$, is used for assessing the worst performance of alternatives by finding the alternative with the minimum maximum deviation from ideal reference values r_i , providing additional robustness to the ranking.

3. 3NAG (Second & Third Normalisation)

a. Second Normalisation (Linear Normalisation): $N_{ij}^2 = \frac{a_{ij}}{\sum a_{ij}}$

b. Third Normalisation (Max Normalisation): $N_{ij}^3 = \frac{a_{ij}}{\max a_{ij}}$

c. Absolute Ordering Calculations:

$$OA_{N1} = A_{ijm} - A_{ijm-\max},$$

$$OA_{N2} = A_{ijm} - A_{ijm-\max},$$

$$OA_{N3} = A_{ijm} - A_{ijm-\max},$$

d. Overall Absolute Ordering (Final Score): $OAG = \sum_{j=i}^m OA_{Nn}$

The 3NAG method provides improvement on the ranking results through multiple normalisation perspectives, which are then combined using an aggregating technique. To start with, the 3NAG method includes two additional normalisation steps, which are then followed by the initial normalisation step of the MOORA method, i.e., vector normalisation. In the second normalisation step, the values are normalised using the linear scaling technique, which divides the values by the sum of the column values, thus providing proportional values that add up to 1. In the third normalisation step, max scaling is used, which scales all the values between 0 and 1, considering the column maximum values. In all normalisation steps, including the initial normalisation step of the MOORA method, an absolute ordering value is

computed as: $OA_{Nm} = A_{ijm} - A_{ijm-max}$, which is essentially the distance of the performance of every alternative from the performance of the best alternative, as determined using a particular normalisation technique. In the 3NAG method, three such distance values are computed using three normalisation techniques, respectively, followed by aggregating all the absolute ordering values into an overall absolute ordering value, which is represented as OAG, as the combined result of all normalisation techniques, thus providing a robust result on ranking alternatives, with higher values of the aggregating variable, i.e., 'O', indicating better performance of alternatives regardless of the normalisation technique used.

2.3.3 CRITIC + EDAS

Methods Used

A novel integrated model by Rasool et al. (2025) used the combination of the CRITIC approach with the EDAS technique, which operates in a LT-SF environment. The CRITIC approach is used for completely objective criterion weight determination based on data. It uses the standard deviation of the contrast intensity of a criterion and its correlations with other criteria. Expert judgment is not necessary for this approach. The EDAS technique is used for decision alternative ranking based on the evaluation of the distance of alternatives from the average solution for a criterion, positive (PDA) or negative (NDA). The LT-SF sets, which have been enriched by the use of the Hamacher operators, are used for capturing the uncertainty, vagueness, or ambiguity associated with expert judgment, for example, "good," "poor," etc.

This methodology is carried out over five phases for green technology selection. Phase 1, Expert Evaluation and Data Collection, Phase 2, Data Processing and Aggregation, Phase 3: CRITIC - Automatic Weighting, Phase 4: EDAS - Alternative Ranking, and Phase 5: Selection. In phase 1, "Expert Evaluation and Data Collection", experts are employed for evaluating the various alternative technologies using linguistic expressions for different criteria. In phase 2, "Data Processing and Aggregation", the linguistic evaluations are converted to LT-SF numbers and aggregated to obtain a final "Collective Decision Matrix" using LT-SF Hamacher aggregation operators. Phase 3, "CRITIC - Automatic Weighting", here, the LT-SF-CRITIC algorithm is employed for the data obtained from Phase 2. This

methodology calculates the standard deviation and the correlation matrix between the criteria from the LT-SF data and derives the objective criterion weights. Next, phase 4 is "EDAS - Alternative Ranking" and here, the LT-SF-EDAS methodology is employed. The average value (AV) for each criterion for all the alternative technologies is obtained. Then, the "Positive Distance from the Average" (PDA) and "Negative Distance from the Average" (NDA) are obtained for each alternative and criterion. These are weighted and aggregated for each alternative and each criterion and ranked based on the appraisal score (AS) obtained. Finally, phase 5, "Selection" and here, the alternative with the highest appraisal score is selected as the most preferred alternative for the green technology selection problem.

Flowchart

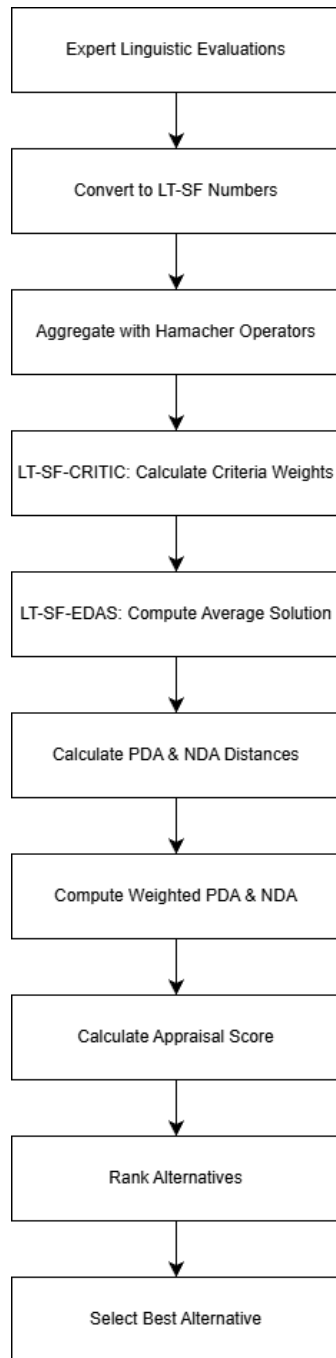


Figure 2.3.3.1: CRITIC–EDAS Methodology Flowchart for Linguistic T-Spherical Fuzzy Decision-Making (Rasool et al., 2025)

The flowchart shows how things work in steps. It starts with Expert Linguistic Evaluations. This is where we do two things which is converted to LT-SF Numbers and Aggregate with Hamacher Operators. After we do these things the information goes two ways at the time. It goes to LT-SF-CRITIC to Calculate Criteria Weights and to LT-SF-EDAS to Compute Average Solution. The LT-SF-EDAS path keeps going with Calculate PDA and NDA Distances. Then Compute Weighted PDA and NDA. This leads to Calculate Appraisal

Scores, for the Expert Linguistic Evaluations. The final steps are "Rank Alternatives" and "Select Best Alternative." This diagram effectively shows the parallel processing of weighting (CRITIC) and ranking setup (EDAS), followed by their integration.

Formula

1. LT-SF-CRITIC Weighting:

- a. Information Measure: $C_j = \sigma_j \sum_{k=1}^n (1 - r_{jk})$
- b. Criterion Weight: $w_j = \frac{C_j}{\sum_{k=1}^n C_k}$

The amount of information C_j contained in criterion j is the product of its standard deviation (σ_j), representing contrast intensity) and the sum of its conflict with other criteria ($1 - r_{jk}$, where r_{jk} is the correlation coefficient). A higher C_j indicates a more important criterion. Weights w_j are obtained by normalizing these information measures.

2. LT-SF-EDAS Ranking:

- a. Average Solution: $AV_j = \frac{\sum_{i=1}^m x_{ij}}{m}$
- b. Positive Distance: $PDA_{ij} = \frac{\max(0, x_{ij} - AV_j)}{AV_j}$
- c. Negative Distance: $NDA_{ij} = \frac{\max(0, AV_j - x_{ij})}{AV_j}$
- d. Weighted Sums: $SP_i = \sum_{j=1}^n w_j \cdot PDA_{ij}$; $SN_i = \sum_{j=1}^n w_j \cdot NDA_{ij}$
- e. Appraisal Score: $AS_i = \frac{1}{2} (NSP_i + (1 - NSN_i))$

The process starts by calculating the average performance AV_j for each criterion. For each alternative i , PDA_{ij} measures its normalized superiority over the average (only if it exceeds the average), while NDA_{ij} measures its normalized inferiority. The weighted sums SP_i and SN_i aggregate these distances. Finally, the appraisal score AS_i is a normalized composite where a higher positive distance and a lower negative distance yield a higher score, determining the final rank.

2.3.4 q-ROF-AHP

Methods Used

A new framework known as a q-ROF-AHP was proposed by Khan et al. (2025). The new framework is a combination of the tree-like ordering of AHP with improved uncertainty management using the enhanced sets described by q-ROFS. There is a hierarchical ordering of the decision problem using AHP, along with subjective weights derived using pair-wise comparison mechanisms. The q-ROFS component significantly extends traditional fuzzy sets by allowing decision-makers to express membership (μ), non-membership (ν), and hesitancy (π) degrees within a broader domain, constrained by the condition $0 \leq (\mu_A(x))^q + \nu_A(x)^q \leq 1$, for $q \geq 1$. One of the major advantages of the proposed approach is the AHP-Based Divergence Distance Measure, which is a consistency test on expert-defined pair-wise comparison matrices using the proposed q-ROF numbers for input verification, followed by weight calculation.

The implementation of this follows a systematic five-phase process. Phase 1: Problem Structuring which involves defining the decision goal, identifying relevant criteria and alternatives and organising them into a traditional AHP hierarchy. In Phase 2: Expert Judgment Collection, domain experts perform pairwise comparisons for elements at each level of the hierarchy. Instead of using crisp numbers, they express their judgments linguistically (e.g., "moderately more important"), which are then converted into q-Rung Orthopair Fuzzy Numbers (q-ROFNs). Phase 3: Consistency Check is critical; the newly proposed divergence distance measure $D(A, B)$ is applied to the matrices of q-ROFNs to evaluate their consistency. Matrices failing to meet a consistency threshold are returned to experts for revision. Phase 4: Weight & Score Calculation proceeds only with consistent matrices. Using eigenvalue methods extended for q-ROFS, the local weights for criteria and the performance scores for alternatives relative to each criterion are computed. Finally, Phase 5: Final Ranking aggregates these local weights and scores through the hierarchy to obtain global priority scores for each alternative, which determine the final ranking.

Flowchart

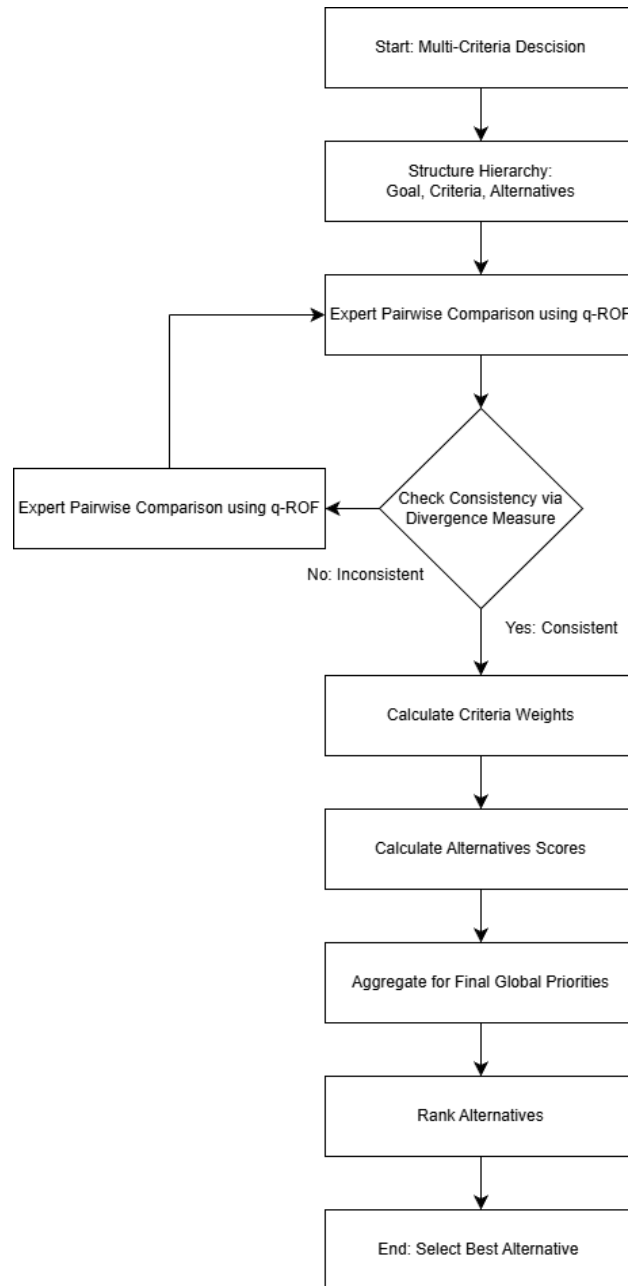


Figure 2.3.4.1: q-ROF–AHP Methodology Flowchart for Fuzzy Multi-Criteria Decision-Making (Khan et al., 2025)

The flowchart commences with Problem Structuring and building a hierarchy. The next block is "Expert Judgment Collection" where linguistic pairs are converted to their respective q-ROFNs. The following block is a decision diamond box titled "Consistency Check using Divergence Measure." If inconsistency is found, we revisit; otherwise, we move to "Calculate Weights and Scores using q-ROF-AHP." This is followed by "Aggregate for Global Priorities and Rank Alternatives," followed by "Final Decision." The flowchart again drives home the need for a feedback loop to ensure consistency in input values.

Formula:

1. q-Rung Orthopair Fuzzy Set (q-ROFS):

- a. Definition: $A = \{(x, \mu_A(x), v_A(x)) \mid x \in X\}$
- b. Constraint: $0 \leq (\mu_A(x))^q + v_A(x)^q \leq 1$, where $q \geq 1$
- c. Hesitancy Degree: $\pi_A(x) = \sqrt[q]{1 - (\mu_A(x))^q - v_A(x)^q}$

This defines a q-ROFS A, where for an element x , $\mu_A(x)$ is its degree of membership, $v_A(x)$ is its degree of non-membership, and $\pi_A(x)$ is the hesitancy degree. The parameter q provides flexibility where a larger q expands the space for μ and v can be assigned, allowing experts to express more nuanced judgments.

2. AHP-based Divergence Distance Measure:

- a. The paper develops a new measure $D(A, B)$ to calculate the distance between two q-ROFSs A and B.
- b. This measure is integrated into the AHP framework to assess the consistency of a pairwise comparison matrix $A = [a_{ij}]$ where each element a_{ij} is a q-ROFN representing the comparison of element i to j . The distance measure quantifies the deviation from perfect consistency, where $a_{ij} \otimes a_{jk}$ should ideally equal a_{ik} for all i, j, k .

This qROFAHP technique is incorporated into the study due to the high level of uncertainty and the subtle judgments that accompany academic advising. With it, the advisor is able to indicate not only the level of which factor of the criteria is more important than the other, but also the level of confidence being attached to the assessment. This adds more realism and transparency to the weighting. Moreover, the technique also checks the logic of the weights given to the experts for the Agentic AI system. This adds to the level of trustworthiness of the recommendation engine.

2.3.5 Relation to Specialisation

The RIASEC model is not only a way of grouping what individuals are attracted to,

career-wise, but it's also a predictive tool in matching personality types with academic and career directions that fit. According to Holland (1997), he claimed that when a person's personality matches the environment, they are usually more contented and successful. Herein, that premise is applied to relate the RIASEC characteristics to the five Computer Science streams offered at KICT.

Below are the 6 RIASEC Traits:

- Investigative (Thinkers):
Individuals who exhibit high Investigative Interest tend to enjoy doing tasks or participating in areas of interest that entail observation, symbolic work, systematic work, and creative exploration of physical, biological, or cultural phenomena (Holland, 1997). These areas of interest are greatly aligned with Artificial Intelligence (AI), Security in Digital Systems (SDS), Data Engineering (DE), as they entail analysis, research, or solving complex problems, as observed with Investigative personalities (Zainudin et al., 2024).
- Realistic (Doers):
Realistic types are attracted to tasks that offer opportunities to do hands-on, systematic, and methodical work with objects, tools, machines, or even animals (Holland, 1997). This makes the trait to become a greater importance for fields such as Network & Data Communication (NDC), Data Engineering (DE), and Application Development Engineering (ADE), as the focus is on constructing or implementing a work in a realistic, hands-on manner (Zainudin et al., 2024).
- Artistic (Creators):
Artistic people generally prefer to engage themselves in work or activities that require less structured or organized involvement, allowing for a scope for exploration, handling tangible, verbal, or human materials and crafting art or artistic products themselves (Holland, 1997). The characteristics can be seen in Application Development Engineering (ADE), particularly in UI/UX and Frontends, and to a lesser degree in Artificial Intelligence, particularly in Creative aspects of AI itself (Wei, 2024).

- **Social (Helpers):**
Social types prefer activities that promote interaction with others, such as informing, teaching, growing, mending, or guiding them (Holland, 1997). Though not a major technical cue, this supports Application Development Engineering (ADE) roles that depend on teamwork, user knowledge, as well as client interactions which is essential to agile software development consultancy (Zainudin et al., 2024).
- **Enterprising (Persuaders):**
Enterprising individuals prefer activities that involve influencing others to attain organisational or economic goals (Holland, 1997). This trait aligns with leadership and business-oriented roles within any specialisation, such as Product Manager or Team Lead in Application Development Engineering (ADE) (Zainudin et al., 2024).
- **Conventional (Organisers):**
Conventional types prefer activities that involve the explicit, ordered, systematic manipulation of data, such as keeping records, filing materials, reproducing materials, organising written and numerical data, and operating data-processing equipment (Holland, 1997). The above has serious implications for data engineering (DE), security in digital systems (SDS), and network & data communication (NDC), most notably through data structuring and adherence to data-driven procedures (Zainudin et al., 2024).

2.5 Summary

This current chapter will delve deeper into four hybrid methods of MCDM (BWM+VIKOR, SWARA+MOORA+3NAG, CRITIC+EDAS, and q-ROF-AHP) and will discuss them as the brain of the novel Agentic AI proposition for adaptive curriculum formation. Before moving into the actual application, we will carefully examine the way each method has been grounded theoretically, through the actual implementation, and mathematically, through actual examples taken from relevant domains such as occupational risk assessment, supplier selection, green technology evaluation or fuzzy decision-making. As it can be seen, hybrid methods of MCDM are highly adaptable tools, ready to confront the

complexity of multi-faceted reality.

However, this chapter develops further than this algorithmic disproof. In fact, it explores how the RIASEC personality types, including Realistic, Investigative, Artistic, Social, Enterprising, and Conventional personalities, can be included as part of the Computer Science specialization advice-making processes. In fact, with RIASEC personalities being linked with Computer Science specializations, recommendation possesses a new dimension, depending on personality types as well as vocation types.

Collectively, results from these studies conducted on MCDM support the effectiveness of this approach to structured decision-making. On the down side, these studies indicate one of the greatest weaknesses of educational advising is the lack of an adaptable, expert-editable MCDM approach, since so much of what is found is static, necessitating rule modifications with developers.

In order to achieve and solve the problem, the chapter suggests the Agentic AI-driven architecture where all the discussed MCDM models will be integrated as parallel reasoners within the flexible architectural setup that can be controlled and adjusted by the academic advisors directly, thus meeting the basic objective of the “agentic.” With the merger of the scientific rigor associated with the MCDM and the adaptive AI models, this chapter manage to tell that the authors have laid the groundwork for the development of the conceptual scale applicable and adaptive academic

CHAPTER THREE

METHODOLOGY

3.1 Research Design

This research employs a quantitative method with a focus on the design and proof of a resultant system. The research can be divided into two clear stages:

- Phase 1 (FYP1): Development of the Agentic AI Engine, along with initial experimentation, working with the Computer Science course curriculum developed by the KICT as a prototype. This consists of objectives such as: design of an adaptive recommendation platform, development of four different types of Hybrid MCDR engines, initial experimentation with historical student record sets.
- Phase 2 (FYP2): Scaling it up to accommodate the universal theme with the help of multiple faculties by widening the structure developed in the database.

The methodology follows an experimental approach, where four MCDM algorithms which is :

1. Bayesian Best-Worst Method (BWM) + ViseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR)
2. Step-Wise Weight Assessment Ratio Analysis (SWARA) + Multi-Objective Optimisation by Ratio Analysis (MOORA) + Three Normalisations and Absolute Ordering (3NAG)
3. Criteria Importance Through Intercriteria Correlation (CRITIC) + Evaluation based on Distance from Average Solution(EDAS) + Linguistic T-Spherical Fuzzy (LT-SF)
4. q-Rung Orthopair Fuzzy (q-ROF) - Analytical Hierarchy Process (AHP)

All the techniques are then tested against the validation benchmarks prepared by experts to identify which mathematical logic acts most like human academic advising. The system is an online-webbed application in which a student, an advisor, and the AI agent can interact.

3.1.1 Flow chart

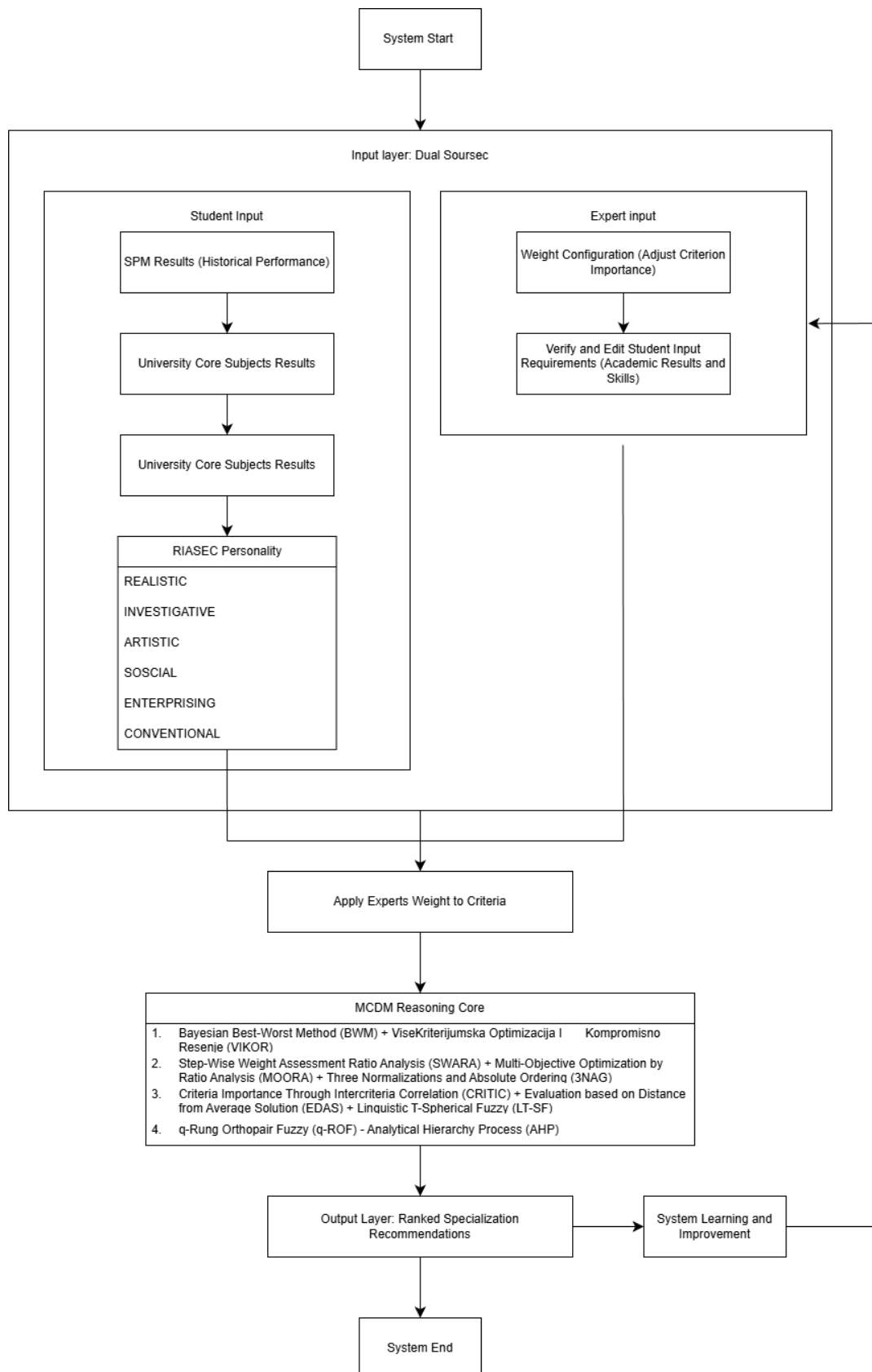


Figure 3.1.1.1: Agentic AI Adaptive Curriculum Formation Flowchart

The figure indicates the Agentic AI-Driven Adaptive Curriculum Recommendation System works end-to-end. The Agentic AI-Driven Adaptive Curriculum Recommendation System begins with a two-data input system. The input comes simultaneously from students and educational advisors. The students provide their educational background, including their outcome for their SPM examination, their grades taken during University courses, their self-assessment of their technical competency level, and their overall RIASEC personality profiling consisting of all six personality types: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional. Simultaneously, educational advisors will have used an expert interface for completing the level of importance for each criterion and validating or correcting student information.

Once data is collected and validated, an operation applies the weighted data to the data collected for students, and then this data passes through the Agentic AI Controller. The controller provides no-code tweaks to decision logic, enabling the main "agentic" feature. The data then passes through the MCDM Reasoning Core, comprising four hybrid approaches for performing MCDM, which all function simultaneously. Hybrid approaches here include BWM + VIKOR, SWARA + MOORA + 3NAG, CRITIC + EDAS, and q-ROF - AHP. Each acts as its own reasoning machine, reliant on its unique mathematical model, and through them all, a list of all applicable specializations is determined. The output data combines in an output layer and provides a final ranking for specializations, along with confidence and rationales for each one.

It culminates in a kind of validation and feedback. Recommendations are compared against expert opinions and past student performance. Discrepancies or new knowledge gleaned may be fed back into the decision-making process for the advisors, allowing them perhaps to adjust their weights or criteria, thus closing the adaptation loop. This flowchart depicts not just the overall method but, more impressively, its potential for expert-in-the-loop adaptation, thereby yielding a roadmap for highly individualized, data-driven student advising.

3.2 Population and Sample:

Below are the population and sample information regarding our project:

- **Population:** The research target group consists of Bachelor of Computer Science (BCS) students at the Kulliyah of Information and Communication Technology (KICT) within the International Islamic University Malaysia (IIUM).
- **Sample:** The sample will be purposive in terms of comprising 10 third and fourth-year students who already have chosen their specialization. This group will cater for different educational backgrounds, varying technical skillsets, and different RIASEC personality profiles. Such data will form a well-rounded foundation for an initial validation of the system.
- **Sampling Method:** This was done to guarantee that the students had finished their decisions in the specialization process, thus being able to draw comparisons between what the system had advocated for and what the students actually chose. Other considerations included the students' accessibility and willingness to take part in the process.

3.3 Data Collection Methods:

Data was collected from multiple structured sources using both digital forms and expert interfaces:

1. Student Profile:

NAME	GENDER	AGE	CURRENT SEMESTER	CURRENT SPECIALISATION
Student 1	Male	22	6	- Artificial Intelligence - Data Engineering
Student 2	Male	22	6	- Artificial Intelligence - Data Engineering
Student 3	Male	23	5	- Networking & Data Communication - Security in Digital System
Student 4	Male	24	7	- Artificial Intelligence - Data Engineering
Student 5	Male	24	6	- Artificial Intelligence - Data Engineering
Student 6	Male	24	6	- Artificial Intelligence - Data Engineering

Student 7	Male	21	5	- Networking & Data Communication - Security in Digital System
Student 8	Male	21	5	- Networking & Data Communication - Security in Digital System
Student 9	Male	22	5	- Artificial Intelligence - Data Engineering
Student 10	Male	22	5	- Networking & Data Communication - Security in Digital System

Table 3.3.1: Student Profile data

2. Student Academic Records:

- SPM Results: Grades were collected for subjects including Modern Mathematics, Additional Mathematics, Physics, Computer Science, English, Chemistry, Biology, Accounting, and Economics.
- University Course Grades (Previous semester results)
 - i. Programming Fundamentals
 - ii. Data Structure and Algorithm
 - iii. Database System, Computer Networking
 - iv. Object Oriented Programming
 - v. Discrete Structures
 - vi. Software Engineering
 - vii. Operating System
 - viii. System Analysis & Design
 - ix. Introduction to Computer Organisation
 - x. ICT & Islam
 - xi. Digital & Embedded Systems
 - xii. Enterprise Networks
 - xiii. Intelligent Systems
 - xiv. Computer Architecture & Assembly Language

- xv. Cross-Platform Software Development
- xvi. Computation and Complexity,
- xvii. User Experience Design.

3. Self-Assessed Technical Skills:

- Students rated their proficiency in 10 technical skills on a 1-5 point scale via Google Form. The skills assessed were:
 - i. Python Programming
 - ii. SQL & Database Management
 - iii. Git Version Control
 - iv. JavaScript / Web Development
 - v. Linux / Command Line
 - vi. Problem Solving Skills
 - vii. Debugging Skills
 - viii. Algorithm Design
 - ix. Shell Scripting
 - x. Cloud Platform (AWS/Azure/GCP)
 - xi. Containerisation (Docker/K8s)

4. RIASEC Personality Assessment:

- An online questionnaire consisting of 30 items (5 items per RIASEC trait) was administered. Students indicated their points on a 1-5 point scale. The items included:
 - i. Realistic (R):
 - 1. "When building furniture, do you enjoy following technical diagrams?"
 - 2. "Do you prefer hands-on activities like fixing electronic devices?"
 - 3. "When playing video games, do you enjoy figuring out game mechanics?"
 - 4. "Have you enjoyed activities involving tools like repairing phones?"
 - 5. "Do you like organising physical spaces for optimal workflow?"
 - ii. Investigative (I):
 - 1. "When watching detective movies, do you try to solve the

- mystery?"
2. "Do you research topics deeply online out of curiosity?"
3. "Have you enjoyed solving complex puzzles like Sudoku?"
4. "When learning, do you prefer understanding the 'why'?"
5. "Do you enjoy analysing patterns in everyday life?"
- iii. Artistic (A):
 1. "Have you designed creative social media posts?"
 2. "Do you enjoy innovative solutions to problems?"
 3. "When decorating, do you focus on unique aesthetics?"
 4. "Have you created original content like digital art?"
 5. "Do you enjoy brainstorming 'outside the box'?"
- iv. Social (S):
 1. "Do you help friends with technology problems?"
 2. "Have you enjoyed teaching concepts to classmates?"
 3. "Do you prefer working on team projects?"
 4. "Have you volunteered to organise events?"
 5. "Do people come to you for advice?"
- v. Enterprising (E):
 1. "Have you taken the lead in organising events?"
 2. "Do you enjoy convincing friends to try new tech?"
 3. "Have you managed a budget for a project?"
 4. "Do you naturally take charge?"
 5. "Have you successfully pitched an idea?"
- vi. Conventional (C):
 1. "Do you enjoy creating organised systems?"
 2. "Have you made detailed plans or checklists?"
 3. "Do you prefer following clear procedures?"
 4. "Have you enjoyed cataloging collections?"
 5. "Do you ensure all details are correct?"
5. Expert Inputs (Criteria Weights):
 - The academic advisor provided criterion weights through three dedicated MCDM configuration interfaces:
 - i. q-ROF-AHP: A pairwise comparison matrix was constructed comparing the four main criteria: SPM Results, Previous Semester

Results, Technical Skills, and Aptitude Test (RIASEC).

- ii. BWM: The Best criterion (Aptitude Test) and Worst criterion (SPM Results) were chosen, followed by importance ratings on a 1–9 scale.
- iii. SWARA: Criteria were ranked, and comparative importance ratios were provided (e.g., Technical Skills are 0.4 less important than Aptitude Test).

6. Tools Used:

- Google Forms for collecting student data.
- Custom web interfaces for expert weight settings.

Four algorithms of MCDM have been implemented, and they are powered by a Python-based backend. The data analysis framework is tailored to conduct an experiment on the Agentic AI system through comprehensive comparisons of the automated recommendations with two ground truths: the decision of the human Academic Advisor and the choices of students on specializations.

3.4 Data Analysis Methods

The analysis will follow through two steps: Algorithmic Processing and Comparative Validation.

3.4.1 Algorithmic Processing and Implementation

A four-solution hybrid MCDM approach sits at the heart and has been developed in Python to match the vectors related to students with relevant criteria in accordance with the expert weighting.

1. Data Normalisation and Processing:

- The raw input variables which is SPM results, university course grades, and technical skill level, are processed to ensure usability.
- Quantitative RIASEC personality traits play a critical part as a dimension in the decision matrix for matching students to suitable specialisations.

2. MCDM Execution:

- There are four independent and parallel reasoning engines that operate: q-ROF-AHP, BWM+VIKOR, SWARA+MOORA-3NAG, and LTSF-CRITIC-EDAS.
- Each will calculate a specialisation ranking list for every student by using the

criteria weights given by the Academic Advisor.

3.4.2 Comparative Validation Strategy

In order to test the relevance and reliability of the recommendations given by the system, the research utilized a “Top-2 Ranking” method. In this method, the basic principle behind a recommendation’s validation lies within the inclusion of the student's actual chosen field or that of the expert's choice within the first two rankings given by the AI.

This choice over a strict "Top-1" comparison was driven by a few practical considerations:

- **Marginal Statistical Differences:** In MCDM calculations, the difference between the first and second choices is relatively tiny and they are effectively interchangeable in regard to suitability.
- **Domain Overlap:** There is so much overlap in core skills between specializations within IT, for instance, Artificial Intelligence and Data Engineering share a lot, meaning a student will be suited for more than one pathway.

Validation Metrics: Individual results of each student were analysed and classified into three categories:

- **Primary Match:** A Rank 1 recommendation by AI matches the real specialty of the student or the expert's pick.
- **Secondary Match:** The AI's Rank 2 recommendation matches the student's actual specialty or the expert's pick.
- **No Match:** Neither of the top two AI recommendations aligns with the benchmark.

Qualitative Analysis: Beyond the question of whether there was a match or not, the study did a qualitative review of Consensus (when both AI and Expert agree) and Divergence (where AI proposes some other choice). This helped reveal instances where Agentic AI might uncover latent talents that human judgment or current student choices might overlook.

3.5 Ethical Considerations

Various ethical considerations guided us through our research process. Most importantly, participants' privacy was given high consideration. The respondents were informed at first of what this research was set out to do, how the researcher planned on doing

it, and what it would accomplish. The participants' data was anonymized such that their ID numbers were changed, and another unique number allocated. The participants' identifiable information is kept different from their research data. Accessing confidential information such as students' records is granted at an institutional level. Lastly, access to data is given only to accredited officials. Transparency was another major thread, where participants were made to understand that this is essentially a recommendation system meant to assist with academic decisions rather than being a predictor of success and ability. As a way of adding integrity to this project, an academic advisor was included to validate the results of this system, with a view to ethics and reliability.

CHAPTER FOUR

RESULTS AND DISCUSSIONS

4.1 Overview of Results

This chapter will discuss the experimental findings gathered from the first phase of the Agentic Artificial Intelligence (AI) Curriculum Recommendation System. The experiment was conducted on a pilot test of 10 undergraduate students from the Kulliyyah of Information and Communication Technology (KICT). The participants were chosen to represent a wide demographic of academic backgrounds (Year 3 and Year 4) and unique personality types as identified by the Realistic, Investigative, Artistic, Social, Enterprising, and Conventional (RIASEC) test.

The information provided below is divided into two main categories:

1. **System-Generated Results:** This part describes the profile of each student (Academic results, Skills, and Personality) and the corresponding rankings produced by the four Agentic Multi-Criteria Decision-Making (MCDM) engines: q-Rung Orthopair Fuzzy Analytic Hierarchy Process (q-ROF-AHP), Best-Worst Method + VlseKriterijumska Optimizacija I Kompromisno Resenje (BWM+VIKOR), Step-wise Weight Assessment Ratio Analysis + Multi-Objective Optimisation - 3 Non-Additive Groups (SWARA+MOORA-3NAG), and Linguistic T-Spherical Fuzzy – Criteria Importance Through Intercriteria Correlation – Evaluation based on Distance
2. **Expert Validation Benchmarks:** This section provides the human rankings given by the KICT Academic Advisor for the same set of students, which is the ground truth for the performance of the system.

4.2 Results

4.2.1 MCDM Rankings

The following tables display the input criteria collected for each student and the

resulting specialisation rankings produced by the four MCDM algorithms.

Student	Student Profile & Assessment Data	MCDM Methods & Rankings			
		q-ROF-AHP	BWM+VIKOR	SWARA+MOORA-3NAG	LTSF-CRITIC-EDAS
1	The detailed student profile and assessment data for Student 1 are presented in Appendix A.1	1. Network & Data Communication 2. Artificial Intelligence 3. Data Engineering 4. Security in Digital System 5. Application Development Engineering	1. Network & Data Communication 2. Security in Digital System 3. Data Engineering 4. Artificial Intelligence 5. Application Development Engineering	1. Network & Data Communication 2. Data Engineering 3. Security in Digital System 4. Application Development Engineering 5. Artificial Intelligence	1. Artificial Intelligence 2. Network & Data Communication 3. Application Development Engineering 4. Security in Digital System 5. Data Engineering
2	The detailed student profile and assessment data for Student 2 are presented in Appendix A.2	1. Network & Data Communication 2. Application Development Engineering 3. Artificial Intelligence 4. Data Engineering 5. Security in Digital System	1. Application Development Engineering 2. Artificial Intelligence 3. Data Engineering 4. Network & Data Communication 5. Security in Digital System	1. Application Development Engineering 2. Artificial Intelligence 3. Data Engineering 4. Network & Data Communication 5. Security in Digital System	1. Data Engineering 2. Artificial Intelligence 3. Network & Data Communication 4. Security in Digital System 5. Application Development Engineering
3	The detailed student profile	1. Network & Data	1. Network & Data	1. Network & Data	1. Network & Data

	and assessment data for Student 3 are presented in Appendix A.3	Communi cation 2. Security in Digital System 3. Applicati on Develop ment Engineeri ng 4. Artificial Intelligen ce 5. Data Engineeri ng	Communi cation 2. Security in Digital System 3. Applicati on Develop ment Engineeri ng 4. Artificial Intelligen ce 5. Data Engineeri ng	Communi cation 2. Security in Digital System 3. Data Engineeri ng 4. Applicati on Develop ment Engineeri ng 5. Artificial Intelligen ce	Communi cation 2. Applicati on Develop ment Engineeri ng 3. Artificial Intelligen ce 4. Security in Digital System 5. Data Engineeri ng
4	The detailed student profile and assessment data for Student 4 are presented in Appendix A.4	1. Artificial Intelligen ce 2. Data Engineeri ng 3. Applicati on Develop ment Engineeri ng 4. Network & Data Communi cation 5. Security in Digital System	1. Artificial Intelligen ce 2. Applicati on Develop ment Engineeri ng 3. Data Engineeri ng 4. Network & Data Communi cation 5. Security in Digital System	1. Artificial Intelligen ce 2. Applicati on Develop ment Engineeri ng 3. Network & Data Communi cation 4. Data Engineeri ng 5. Security in Digital System	1. Artificial Intelligen ce 2. Network & Data Communi cation 3. Data Engineeri ng 4. Applicati on Develop ment Engineeri ng 5. Security in Digital System
5	The detailed student profile and assessment data for Student 5 are presented in Appendix A.5	1. Applicati on Develop ment Engineeri ng 2. Network & Data Communi cation 3. Artificial Intelligen	1. Applicati on Develop ment Engineeri ng 2. Network & Data Communi cation 3. Security in Digital	1. Network & Data Communi cation 2. Applicati on Develop ment Engineeri ng 3. Artificial Intelligen	1. Network & Data Communi cation 2. Applicati on Develop ment Engineeri ng 3. Artificial Intelligen

		4. Security in Digital System 5. Data Engineering	4. Artificial Intelligence 5. Data Engineering	4. Security in Digital System 5. Data Engineering	4. Security in Digital System 5. Data Engineering
6	The detailed student profile and assessment data for Student 6 are presented in Appendix A.6	1. Application Development Engineering 2. Security in Digital System 3. Network & Data Communication 4. Data Engineering 5. Artificial Intelligence	1. Application Development Engineering 2. Security in Digital System 3. Data Engineering 4. Network & Data Communication 5. Artificial Intelligence	1. Application Development Engineering 2. Security in Digital System 3. Data Engineering 4. Network & Data Communication 5. Artificial Intelligence	1. Application Development Engineering 2. Network & Data Communication 3. Artificial Intelligence 4. Security in Digital System 5. Data Engineering
7	The detailed student profile and assessment data for Student 7 are presented in Appendix A.7	1. Application Development Engineering 2. Artificial Intelligence 3. Network & Data Communication 4. Data Engineering 5. Security in Digital System	1. Application Development Engineering 2. Network & Data Communication 3. Artificial Intelligence 4. Data Engineering 5. Security in Digital System	1. Application Development Engineering 2. Network & Data Communication 3. Artificial Intelligence 4. Data Engineering 5. Security in Digital System	1. Network & Data Communication 2. Application Development Engineering 3. Data Engineering 4. Artificial Intelligence 5. Security in Digital System
8	The detailed student profile	1. Network & Data	1. Network & Data	1. Network & Data	1. Network & Data

	and assessment data for Student 8 are presented in Appendix A.8	Communication 2. Artificial Intelligence 3. Security in Digital System 4. Application Development Engineering 5. Data Engineering	Communication 2. Security in Digital System 3. Application Development Engineering 4. Artificial Intelligence 5. Data Engineering	Communication 2. Security in Digital System 3. Data Engineering 4. Application Development Engineering 5. Artificial Intelligence	Communication 2. Artificial Intelligence 3. Application Development Engineering 4. Security in Digital System 5. Data Engineering
9	The detailed student profile and assessment data for Student 9 are presented in Appendix A.9	1. Artificial Intelligence 2. Application Development Engineering 3. Network & Data Communication 4. Data Engineering 5. Security in Digital System	1. Artificial Intelligence 2. Application Development Engineering 3. Network & Data Communication 4. Data Engineering 5. Security in Digital System	1. Artificial Intelligence 2. Application Development Engineering 3. Network & Data Communication 4. Data Engineering 5. Security in Digital System	1. Artificial Intelligence 2. Data Engineering 3. Security in Digital System 4. Network & Data Communication 5. Application Development Engineering
10	The detailed student profile and assessment data for Student 10 are presented in Appendix A.10	1. Network & Data Communication 2. Security in Digital System 3. Artificial Intelligence 4. Data Engineering	1. Network & Data Communication 2. Security in Digital System 3. Data Engineering 4. Artificial Intelligence	1. Network & Data Communication 2. Security in Digital System 3. Data Engineering 4. Application	1. Security in Digital System 2. Network & Data Communication 3. Application Development Engineering

		ng 5. Applicati on Develop ment Engineeri ng	ce 5. Applicati on Develop ment Engineeri ng	Develop ment Engineeri ng 5. Artificial Intelligen ce	ng 4. Artificial Intelligen ce 5. Data Engineeri ng
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Table 4.2.1: Student Profile, Assessment Data, and MCDM Rankings for Every Students

4.2.2 Expert Validation Rankings

To validate the outputs generated by the Agentic AI, expert judgment was obtained from the KICT Academic Advisor. She provided manual, expert-derived rankings for each of the 10 student profiles. These rankings serve as the ground truth benchmark for evaluating the performance and accuracy of the proposed system. Below are the rankings provided:

Student	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
Student 1	Security in Digital System	Data Engineering	Artificial Intelligence	Application Development Engineering	Network & Data Communication
Student 2	Application Development Engineering	Artificial Intelligence	Data Engineering	Security in Digital System	Network & Data Communication
Student 3	Network & Data Communication	Security in Digital System	Application Development Engineering	Artificial Intelligence	Data Engineering
Student 4	Network & Data Communication	Security in Digital System	Application Development Engineering	Artificial Intelligence	Data Engineering
Student 5	Application Development Engineering	Network & Data Communication	Security in Digital System	Artificial Intelligence	Data Engineering
Student 6	Network &	Security in	Data	Artificial	Application

	Data Communicati on	Digital System	Engineering	Intelligence	Development Engineering
Student 7	Security in Digital System	Artificial Intelligence	Data Engineering	Application Development Engineering	Network & Data Communicati on
Student 8	Application Development Engineering	Security in Digital System	Network & Data Communicati on	Artificial Intelligence	Data Engineering
Student 9	Artificial Intelligence	Application Development Engineering	Security in Digital System	Network & Data Communicati on	Data Engineering
Student 10	Network & Data Communicati on	Security in Digital System	Application Development Engineering	Artificial Intelligence	Data Engineering

Table 4.2.2: Expert (KICT Academic Advisor) Rankings for All 10 Students

4.3 Discussion

To better test the relevance of the Agentic AI system, this section proceeds from validation towards more practical testing by making a comparison between the specialisation advice produced by the system, the Academic Advisor, and the current specialisation being followed by the students.

For this analysis, the "Top-2 Ranking" method is employed, and accuracy is measured based on whether the true specialisation of the student is among the top two positions in the output recommended. This method is selected instead of "Top-1" ranking for several very important reasons:

1. Mitigating Marginal Statistical Differences: In Multi-Criteria Decision

Making, the mathematical marginal difference between the ranking of the first alternative and that of the second is trivial (for instance, 85.2% versus 84.8%). In Top-1 decision making, a marginal difference could be considered punishable by penalization for placing a good specialisation in the second rank. Top-2 takes this into consideration.

2. Accommodating Domain Overlap: There are cases where overlapping skills in the domain of various tech fields exist. Consider, for example, the field of Artificial Intelligence and Data Engineering. These two require the same set of skills in mathematics and computer programming. The student could be equally qualified for both. Hence, the difference in Rank 1 and Rank 2 would depend solely upon his preference. The window needs to be expanded to include the top two choices.
3. Balancing Precision and Focus: While it may be true that adding more comparisons for Top 3 or more would yield a better “hit rate”, it would compromise the level of precision for the recommendation system, as a Top 3 recommendation covers 60% of the possible choices, and thereby, it is not a challenging recommendation. By restricting it to Top 2, it retains a high level of precision with some flexibility for human error and known margins of error for statistics.

4.3.1 Analysis of Result

1. Analysis of result for Student 1

Current	Recommendation Source & Top 2 Rankings	Validation Result
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Specialisation			
- Artificial Intelligence - Data Engineering	Expert (Academic Advisor)	1. Security in Digital System 2. Data Engineering	✓ Secondary Match
	q-ROF- AHP	1. Network & Data Communication 2. Artificial Intelligence	✓ Secondary Match
	BWM+ VIKOR	1. Network & Data Communication 2. Security in Digital System	✗ No Match with Student's Current Specialization
	SWARA+ MOORA- 3NAG	1. Network & Data Communication 2. Data Engineering	✓ Secondary Match
	LTSF- CRITIC- EDAS	1. Artificial Intelligence 2. Network & Data Communication	✓ Primary Match

Table 4.3.1.1: Validation of Expert and System Rankings against Current Specialisation for Student 1

- The LTSF-CRITIC-EDAS method proved to be the most accurate model, achieving a Primary Match by correctly identifying Artificial Intelligence as the Rank 1 recommendation. This indicates a strong alignment between the student's objective data and the requirements for the Artificial Intelligence field.
- In comparison, the human Expert Judgment (Academic Advisor) achieved only a Secondary Match, while it correctly identified Data Engineering at Rank 2, it prioritized Security in Digital Systems which is a field not pursued by the student as the primary choice.
- Both q-ROF-AHP and SWARA+MOORA-3NAG successfully achieved Secondary Matches too, identifying Artificial Intelligence and Data Engineering at Rank 2, respectively.
- The BWM+VIKOR method resulted in a No Match, as it recommended Network &

Data Communication and Security, failing to capture either of the student's actual specialisations in its top rankings.

2. Analysis of result for Student 2

Current Specialisation	Recommendation Source & Top 2 Rankings		Validation Result
- Artificial Intelligence - Data Engineering	Expert (Academic Advisor)	1. Application Development Engineering 2. Artificial Intelligence	✓ Secondary Match
	q-ROF- AHP	1. Network & Data Communication 2. Application Development Engineering	✗ No Match with Student's Current Specialization
	BWM+ VIKOR	1. Application Development Engineering 2. Artificial Intelligence	✓ Secondary Match
	SWARA+ MOORA- 3NAG	1. Application Development Engineering 2. Artificial Intelligence	✓ Secondary Match
	LTSF- CRITIC- EDAS	1. Data Engineering 2. Artificial Intelligence	✓ Primary Match ✓ Secondary Match

Table 4.3.1.2: Validation of Expert and System Rankings against Current Specialisation for Student 2

- The LTSF-CRITIC-EDAS model demonstrated the highest alignment accuracy, achieving a Primary Match by correctly identifying Data Engineering as the Rank 1 recommendation. Furthermore, it successfully captured the student's dual-domain focus by placing Artificial Intelligence at Rank 2, making it the only method to

validate both aspects of the student's current specialisation in the top rankings.

- A strong consensus existed between the human Expert Judgment and the agentic models BWM+VIKOR and SWARA+MOORA-3NAG. All three sources independently reached Secondary Matches, accurately pinpointing the Rank 2 position for Artificial Intelligence. However, they each uniformly placed Application Development Engineering in the Rank 1 position, which was not one of the fields the student chose.
- In the q-ROF-AHP method, the outcome was No Match since it suggested Network & Data Communication and Application Development Engineering. This implies a significant deviation from the student's real academic strand in Data Engineering and Artificial Intelligence for this particular model.

3. Analysis of result for Student 3

Current Specialisation	Recommendation Source & Top 2 Rankings		Validation Result
- Network & Data Communication - Security in Digital System	Expert (Academic Advisor)	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match
	q-ROF- AHP	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match
	BWM+ VIKOR	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match
	SWARA+ MOORA- 3NAG	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match
	LTSF- CRITIC- EDAS	1. Network & Data Communication 2. Application Development	✓ Primary Match

		Engineering	
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Table 4.3.1.3: Validation of Expert and System Rankings against Current Specialisation for Student 3

- A high degree of accuracy has been attained for Student 3, for whom Expert Judgment and all but one Agentic AI approaches (q-ROF-AHP, BWM+VIKOR, and SWARA+MOORA-3NAG) were in agreement. Indeed, all four methods attained Primary and Secondary Matches, for whom Rank 1 is Network & Data Communication, followed by Rank 2 for Security in Digital Systems.
- The system was able to properly verify the student's complete academic background. The system's accuracy in predicting both the primary specialization (Network & Data Communication) as well as the secondary specialization (Security in Digital System) using diverse prediction techniques reflected well-established integrity in dealing with students who have expertise in both areas.
- The LTSF-CRITIC-EDAS method again successfully reached a Primary Match with Network & Data Communication at Rank 1. Although a minor deviation in the secondary ranking occurred when they recommended Application Development Engineering instead of the real second specialisation of the student, which is 'Security in Digital System'.

4. Analysis of result for Student 4

Current Specialisation	Recommendation Source & Top 2 Rankings		Validation Result
- Artificial Intelligence - Data Engineering	Expert (Academic Advisor)	1. Network & Data Communication 2. Security in Digital System	✗ No Match with Student's Current Specialization
	q-ROF- AHP	1. Artificial Intelligence 2. Data Engineering	✓ Primary Match ✓ Secondary Match
	BWM+ VIKOR	1. Artificial	✓ Primary Match

		Intelligence 2. Application Development Engineering	
	SWARA+ MOORA- 3NAG	1. Artificial Intelligence 2. Application Development Engineering	✓ Primary Match
	LTSF- CRITIC- EDAS	1. Artificial Intelligence 2. Network & Data Communication	✓ Primary Match

Table 4.3.1.4: Validation of Expert and System Rankings against Current Specialisation for Student 4

- Remarkably, there was an obvious consensus among all four Agentic AI methods (q-ROF-AHP, BWM+VIKOR, SWARA+MOORA-3NAG, and LTSF-CRITIC-EDAS). All system models obtained the Primary Match, and they all declared Artificial Intelligence to be the Rank 1 choice. This depicts the great reliability of the system regarding the identification of the student's
- The q-ROF-AHP method recorded the greatest level of precision amongst all sources. The engine achieved both the Primary and the Secondary Matches by ranking the fields of Artificial Intelligence first and Data Engineering second, thus reflecting the student's twin domain specialization.
- In contrast to the system's success, the human Expert Judgment resulted in a No Match. The advisor recommended Network & Data Communication and Security in Digital Systems, failing to identify either of the student's actual specialisations in the top rankings. This highlights the system's capability to correct potential human oversight regarding the student's suitability for the Artificial Intelligence and Data Engineering fields.

5. Analysis of result for Student 5

Current Specialisation	Recommendation Source & Top 2 Rankings		Validation Result
- Artificial Intelligence - Data Engineering	Expert (Academic Advisor)	1. Application Development Engineering 2. Network & Data Communication	✗ No Match with Student's Current Specialization
	q-ROF- AHP	1. Application Development Engineering 2. Network & Data Communication	✗ No Match with Student's Current Specialization
	BWM+ VIKOR	1. Application Development Engineering 2. Network & Data Communication	✗ No Match with Student's Current Specialization
	SWARA+ MOORA- 3NAG	1. Network & Data Communication 2. Application Development Engineering	✗ No Match with Student's Current Specialization
	LTSF- CRITIC- EDAS	1. Network & Data Communication 2. Application Development Engineering	✗ No Match with Student's Current Specialization

Table 4.3.1.5: Validation of Expert and System Rankings against Current Specialisation for Student 5

- A significant finding for Student 5 is the complete divergence across all decision-making sources. Neither the human Expert Judgment nor any of the four Agentic Artificial Intelligence methodologies achieved a match with the student's current dual specialisation in Artificial Intelligence or Data Engineering within the Top 2 rankings.
- Nonetheless, there has been a strong majority consensus between the Expert and all four MCDM techniques (q-ROF-AHP, BWM+VIKOR, SWARA+ MOORA-3NAG, and finally, the LTSF-CRITIC-EDAS), where all the sources rank Application Development Engineering and then Network & Data Communication more frequently

among the most important, implying that the student's educational background and abilities are more statistically related to either of those disciplines compared to the branch they are following now.

- The steady "No Match" outcome draws attention to a possible mismatch between the student's interest and aptitude. Both the system and the expert found that the student has an implicit fit to Application Development Engineering or Networking and Data Communication, suggesting that the student's current specialization allocation might be impacted by some external consideration instead of the performance measures covered by the system.

6. Analysis of result for Student 6

Current Specialisation	Recommendation Source & Top 2 Rankings		Validation Result
- Artificial Intelligence - Data Engineering	Expert (Academic Advisor)	1. Network & Data Communication 2. Security in Digital System	✗ No Match with Student's Current Specialization
	q-ROF- AHP	1. Application Development Engineering 2. Security in Digital System	✗ No Match with Student's Current Specialization
	BWM+ VIKOR	1. Application Development Engineering 2. Security in Digital System	✗ No Match with Student's Current Specialization
	SWARA+ MOORA- 3NAG	1. Application Development Engineering 2. Security in Digital System	✗ No Match with Student's Current Specialization
	LTSF- CRITIC- EDAS	1. Application Development Engineering 2. Network & Data Communication	✗ No Match with Student's Current Specialization

Table 4.3.1.6: Validation of Expert and System Rankings against Current Specialisation for Student 6

- A complete divergence was observed across all decision-making sources for Student 6. Neither the human Expert Judgment nor any of the four Agentic AI methodologies achieved a match with the student's current dual specialisation in Artificial Intelligence or Data Engineering within the Top 2 rankings.
- While not aligned with the student's chosen path, a great consensus existed among the Agentic AI models in the form of q-ROF-AHP, BWM+VIKOR, SWARA+MOORA-3NAG, and LTSF-CRITIC-EDAS. These sources have constantly recommended Application Development Engineering as the Rank 1 recommendation; therefore, a statistical fit of the student's profile to this field has been maintained by these sources.
- Both Expert Judgment and three system models' recommendations/q-ROF-AHP, BWM+VIKOR, SWARA+MOORA-3NAG positioned the Rank 2 recommendation as Security in Digital Systems. This strengthens the finding that the student indeed has latent aptitude for Security in Digital System roles, even if these are different from his current academic pursuit.

7. Analysis of result for Student 7

Current Specialisation	Recommendation Source & Top 2 Rankings		Validation Result
- Network & Data Communication - Security in Digital System	Expert (Academic Advisor)	1. Security in Digital System 2. Artificial Intelligence	✓ Primary Match
	q-ROF- AHP	1. Application Development Engineering 2. Artificial Intelligence	✗ No Match with Student's Current Specialization
	BWM+ VIKOR	1. Application Development	✓ Secondary Match

		Engineering 2. Network & Data Communication	
	SWARA+ MOORA- 3NAG	1. Application Development Engineering 2. Network & Data Communication	✓ Secondary Match
	LTSF- CRITIC-EDAS	1. Network & Data Communication 2. Application Development Engineering	✓ Primary Match

Table 4.3.1.7: Validation of Expert and System Rankings against Current Specialisation for Student 7

- There was a divergence with success in the human Expert Judgment when compared with the objective model of LTSF-CRITIC-EDAS. Both models were able to attain Primary Matches, although both validated different elements of the dual field of specialization of the student. The human Expert validated Security in Digital Systems with Rank 1, while the model CRITIC validated Network & Data Communication with Rank 1. Taken together, it suggests that the student's profile is equally suited for both options.
- The BWM+VIKOR & SWARA+MOORA-3NAG engines applied consistent algorithmic logic, both achieving Secondary Matches. Although these models rated Application Development Engineering at the primary position, they appropriately identified the student skills for Network & Data Communication at the secondary position.
- The result for q-ROF-AHP was a No Match, suggesting Application Development Engineering and Artificial Intelligence as alternatives. The result shows that this particular model has failed to reflect the student's skills shown for Networking and Security among its preferred choices.

8. Analysis of result for Student 8

Current Specialisation	Recommendation Source & Top 2 Rankings		Validation Result
- Network & Data Communication - Security in Digital System	Expert (Academic Advisor)	1. Application Development Engineering 2. Security in Digital System	✓ Secondary Match
	q-ROF- AHP	1. Network & Data Communication 2. Artificial Intelligence	✓ Primary Match
	BWM+ VIKOR	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match
	SWARA+ MOORA- 3NAG	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match
	LTSF- CRITIC- EDAS	1. Network & Data Communication 2. Artificial Intelligence	✓ Primary Match

Table 4.3.1.8: Validation of Expert and System Rankings against Current Specialisation for Student 8

- There was a strong consensus on all four Agentic AI models. The models, namely q-ROF-AHP, BWM+VIKOR, SWARA+MOORA-3NAG, and LTSF-CRITIC-EDAS, all provided Primary Matches with Network & Data Communication being their Rank 1 choice. This goes on to establish the reliability factor of the system with respect to the primary academic qualification of the student.
- Both the models of BWM+VIKOR & SWARA+MOORA-3NAG performed well by proving the positional correctness of the student's profile. Both were able to achieve the Primary & Secondary Matches accurately, where Network & Data Communication ranked at position 1, followed by Security in Digital Systems at rank 2.
- Although Expert Judgment correctly identified Secondary Match for Security in

Digital Systems at Rank 2, they incorrectly identified the student's main Networking specialisation in preference to Application Development Engineering. Similarly, q-ROF-AHP & LTSF-CRITIC-EDAS incorrectly identified Application Development Engineering in preference to Application Development Engineering for the main Networking specialisation but incorrectly identified Secondary Match at Artificial Intelligence instead of Security in Digital Systems.

9. Analysis of result for Student 9

Current Specialisation	Recommendation Source & Top 2 Rankings		Validation Result
- Artificial Intelligence - Data Engineering	Expert (Academic Advisor)	1. Artificial Intelligence 2. Application Development Engineering	✓ Primary Match
	q-ROF- AHP	1. Artificial Intelligence 2. Application Development Engineering	✓ Primary Match
	BWM+ VIKOR	1. Artificial Intelligence 2. Application Development Engineering	✓ Primary Match
	SWARA+ MOORA- 3NAG	1. Artificial Intelligence 2. Application Development Engineering	✓ Primary Match
	LTSF- CRITIC- EDAS	1. Artificial Intelligence 2. Data Engineering	✓ Primary Match ✓ Secondary Match

Table 4.3.1.9: Validation of Expert and System Rankings against Current Specialisation for Student 9

- A strong consensus emerged in each of the four Agentic AI techniques as well as in Expert Judgment. The engines q-ROF-AHP, BWM+VIKOR, SWARA+MOORA-3NAG, as well as LTSF-CRITIC-EDAS, as well as Academic Advisor, collectively pinpointed Artificial Intelligence as the Rank 1 suggestion. Notably, this attests to the high reliability of this system in identifying the scholar's underlying academic talent
- Modeling-wise, the LTSF-CRITIC-EDAS model proved its efficiency in verifying the student's profile. In only the second model, the Primary Match and Secondary Match were accomplished with accuracy, positioning Artificial Intelligence at Position 1 and Data Engineering at Position 2.
- Although the Expert Judgment, alongside the three algorithmic approaches (q-ROF-AHP, BWM+VIKOR, and SWARA+MOORA-3NAG), was successful in ranking the core Artificial Intelligence specialisation correctly, the quartet failed to follow the student's actual secondary specialisation. This is despite the four ranking the secondary specialisation, Application Development Engineering, at the correct position of Rank 2, having overlooked the student's secondary specialisation in Data Engineering, which was accurately determined by the LTSF-CRITIC-EDAS

10. Analysis of result for Student 10

Current Specialisation	Recommendation Source & Top 2 Rankings		Validation Result
- Network & Data Communication - Security in Digital System	Expert (Academic Advisor)	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match
	q-ROF- AHP	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match
	BWM+ VIKOR	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match

	SWARA+ MOORA- 3NAG	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match
	LTSF- CRITIC- EDAS	1. Network & Data Communication 2. Security in Digital System	✓ Primary Match ✓ Secondary Match

Table 4.3.1.10: Validation of Expert and System Rankings against Current Specialisation for Student 10

- A perfect accuracy was witnessed for each of the four Agentic AI approaches. The q-ROF-AHP, BWM+VIKOR, SWARA+MOORA-3NAG, and LTSF-CRITIC-EDAS approaches made Primary Match scores as Network & Data Communication for Rank 1, thereby clarifying that it is a highly reliable system for identifying the main subject competence of a student consistently.
- All four algorithmic models showed improved performance by being able to validate the entire dual specialisation portfolio of the student. Each algorithm showed success in achieving both the Primary and Secondary Matches, where they accurately ranked Network & Data Communication at Rank 1 and Security in Digital Systems at Rank 2, which symbolized a situation of 100% accuracy among all the computational models.
- Unlike those involving subjective variation, the Expert Judgment was consistent with the algorithmic result. The Academic Advisor was able to successfully identify both the primary and secondary specializations (Network & Data Communication and Security in Digital Systems, respectively), resulting in full consistency between the expert judgment and objective algorithmic judgment.

4.4 Conclusion

This research work has successfully demonstrated how the Agentic AI integrated curriculum recommendation system accomplishes the task of converting the static nature of academic advising into a dynamic and expert-led process. This system further accomplishes

this act through the amalgamation of the effects of four different Hybrid Multi-Criteria Decision Making algorithms in conjunction—namely q-ROF-AHP, BWM-VIKOR, SWARA-MOORA-3NAG, and LTSF-CRITIC-EDAS.

The findings obtained from the experiment conducted among 10 students at KICT, IIUM, raise several salient findings:

- **High Algorithmic Reliability:** There was excellent consensus achievement with expert judgment, especially via the LTSF-CRITIC-EDAS and q-ROF-AHP model, which often obtained the “Primary Matches” by pinpointing the students’ real specializations as the best suggestion.
- **Identification of Undiscovered Ability:** In instances where students made different decisions from not only the AI decisions but also the expert decisions (for example, Students 5 and 6), the system pointed out possible conflicting interests. This was in relation to the discovery of hidden technical suitability that would not be possible by conventional personal choice.
- **Agentic Flexibility:** The system satisfied its primary task of enabling academic counselors to change decision factors and weights without any modification to the backend system, thus filling the gap between fixed tools and the needs of the changing education sector.

Ultimately, the project validates the fact that hybrid MCDM solutions are capable of efficiently handling vast dimensions of data such as the 4 key criteria (SPM results, academic grades in universities, technical skills, and RIASEC personality types).

4.5 Future Works

Based on the successful validation of the Phase 1 prototype, and considering a few more aspects, the following directions are suggested for Phase 2:

- **Universal Scalability:** Expand the system's database structure to accommodate multi-faculty data schemas. This will allow the system to function as a "Universal Academic Advisor" capable of supporting different faculties/kulliyyah (Engineering, Economics) simply by inputting their specific criteria.
- **Integration of Predictive Analytics:** Provide historical student outcome data (such

as job placement after graduation) fed into the feedback loop to further fine-tune predictive accuracy in recommendations.

- **Improved User Interface:** Implementing easier-to-navigate dashboards for both academic advisors and students will make the user experience more intuitive.

APPENDICES

A. STUDENT PROFILE AND ASSESSMENT DATA

A.1 Student 1

GENDER		AGE		CURRENT SEMESTER		CURRENT SPECIALISATION	
Male		22		6		- Artificial Intelligence - Data Engineering	
SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A	Programming Fundamentals	A-	Python Programming	6	Realistic	21
Additional Mathematics	B+	Data Structure and Algorithm	A	SQL & Database management	2	Investigative	21
Physics	C+	Database System	C+	Git Version Control	2	Artistic	15
Computer Science		Computer Networking	A	JavaScript / Web Development	6	Social	16
English	B+	Object Oriented Programming	B+	Linux / Command Line	5	Enterprising	17
Chemistry	B+	Discrete Structures		Problem Solving Skills	7	Conventional	17
Biology	C+	Software Engineering	A	Debugging Skills	6		
Accounting		Operating System	B+	Algorithm Design	4		
Economics		System Analysis & Design		Shell Scripting	3		

	Introduction to Computer Organisation		Cloud Platform (AWS/Azure/GCP)	5	
	ICT & Islam	B+	Containerisation (Docker/K8s)	1	
	Digital & Embedded Systems				
	Enterprise Networks				
	Intelligent Systems				
	Computer Architecture & Assembly Language	A			
	Cross-Platform Software Development				
	Computation and Complexity				
	User Experience Design				

Table A.1: Student Profile and Assessment Data of Student 1

A.2 Student 2

GENDER	AGE	CURRENT SEMESTER	CURRENT SPECIALISATION
Male	22	6	- Artificial Intelligence - Data Engineering

SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A	Programming Fundamentals	A-	Python Programming	8	Realistic	14
Additional Mathematics	A-	Data Structure and Algorithm	A	SQL & Database management	7	Investigative	20
Physics	B+	Database System	A	Git Version Control	9	Artistic	24
Computer Science		Computer Networking	A-	JavaScript / Web Development	10	Social	20
English	A	Object Oriented Programming	A	Linux / Command Line	10	Enterprising	25
Chemistry	B+	Discrete Structures	A-	Problem Solving Skills	10	Conventional	25
Biology	B+	Software Engineering	A-	Debugging Skills	10		
Accounting		Operating System	A-	Algorithm Design	8		
Economics		System Analysis & Design	A	Shell Scripting	10		
		Introduction to Computer Organisation	A	Cloud Platform (AWS/Azure/GCP)	1		
		ICT & Islam	A	Containerisation (Docker/K8s)	8		
		Digital & Embedded Systems					
		Enterprise Networks					
		Intelligent Systems	B+				

	Computer Architecture & Assembly Language	B+		
	Cross-Platform Software Development			
	Computation and Complexity			
	User Experience Design	A-		

Table A.2: Student Profile and Assessment Data of Student 2

A.3 Student 3

GENDER		AGE		CURRENT SEMESTER		CURRENT SPECIALISATION	
Male		23		5		- Networking & Data Communication - Security in Digital System	
SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A	Programming Fundamentals	A	Python Programming	4	Realistic	21
Additional Mathematics	C+	Data Structure and Algorithm	A	SQL & Database management	6	Investigative	18
Physics		Database System	B+	Git Version Control	3	Artistic	16
Computer		Computer	A	JavaScript /	7	Social	17

Science		Networking		Web Development			
English	C+	Object Oriented Programming	B+	Linux / Command Line	5	Enterprising	19
Chemistry		Discrete Structures		Problem Solving Skills	6	Conventional	19
Biology		Software Engineering	B	Debugging Skills	6		
Accounting	D	Operating System	A-	Algorithm Design	6		
Economics	A	System Analysis & Design	B+	Shell Scripting	4		
		Introduction to Computer Organisation	A	Cloud Platform (AWS/Azure/GCP)	6		
		ICT & Islam		Containerisation (Docker/K8s)	5		
		Digital & Embedded Systems	A				
		Enterprise Networks					
		Intelligent Systems	A-				
		Computer Architecture & Assembly Language	A-				
		Cross-Platform Software Development					
		Computation and Complexity					
		User Experience					

	Design			
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Table A.3: Student Profile and Assessment Data of Student 3

A.4 Student 4

GENDER		AGE		CURRENT SEMESTER		CURRENT SPECIALISATION	
Male		24		7		- Artificial Intelligence - Data Engineering	
SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A	Programming Fundamentals	A-	Python Programming	5	Realistic	23
Additional Mathematics	B	Data Structure and Algorithm	A-	SQL & Database management	4	Investigative	17
Physics	C	Database System	A-	Git Version Control	3	Artistic	15
Computer Science		Computer Networking		JavaScript / Web Development	3	Social	19
English	B	Object Oriented Programming	A	Linux / Command Line	2	Enterprising	17
Chemistry	D	Discrete Structures		Problem Solving Skills	7	Conventional	24
Biology		Software Engineering	A-	Debugging Skills	5		
Accounting		Operating System		Algorithm Design	5		
Economics		System		Shell Scripting	2		

		Analysis & Design				
		Introduction to Computer Organisation	A	Cloud Platform (AWS/Azure/GCP)	1	
		ICT & Islam	A	Containerisation (Docker/K8s)	1	
		Digital & Embedded Systems				
		Enterprise Networks				
		Intelligent Systems	B+			
		Computer Architecture & Assembly Language	B+			
		Cross-Platform Software Development				
		Computation and Complexity				
		User Experience Design				

Table A.4: Student Profile and Assessment Data of Student 4

A.5 Student 5

GENDER	AGE	CURRENT SEMESTER	CURRENT SPECIALISATION
Male	24	6	- Artificial Intelligence - Data Engineering

SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A+	Programming Fundamentals	A	Python Programming	6	Realistic	22
Additional Mathematics	A-	Data Structure and Algorithm	A	SQL & Database management	6	Investigative	19
Physics	A	Database System	A	Git Version Control	3	Artistic	12
Computer Science		Computer Networking	A	JavaScript / Web Development	7	Social	20
English	A-	Object Oriented Programming	A	Linux / Command Line	5	Enterprising	20
Chemistry	A-	Discrete Structures		Problem Solving Skills	8	Conventional	16
Biology	A+	Software Engineering	A	Debugging Skills	3		
Accounting		Operating System	A-	Algorithm Design	6		
Economics		System Analysis & Design	A	Shell Scripting	3		
		Introduction to Computer Organisation	A	Cloud Platform (AWS/Azure/GCP)	1		
		ICT & Islam	A-	Containerisation (Docker/K8s)	1		
		Digital & Embedded Systems					
		Enterprise Networks					
		Intelligent	A-				

	Systems			
	Computer Architecture & Assembly Language	A		
	Cross-Platform Software Development			
	Computation and Complexity			
	User Experience Design			

Table A.5: Student Profile and Assessment Data of Student 5

A.6 Student 6

GENDER		AGE		CURRENT SEMESTER		CURRENT SPECIALISATION	
Male		24		6		- Artificial Intelligence - Data Engineering	
SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A+	Programming Fundamentals	A-	Python Programming	5	Realistic	23
Additional Mathematics	A	Data Structure and Algorithm	B-	SQL & Database management	7	Investigative	15
Physics	A	Database System	B+	Git Version Control	2	Artistic	17
Computer Science		Computer Networking	A	JavaScript / Web	4	Social	14

				Development			
English	A	Object Oriented Programming	B+	Linux / Command Line	2	Enterprising	19
Chemistry	A	Discrete Structures		Problem Solving Skills	7	Conventional	22
Biology	A-	Software Engineering	A-	Debugging Skills	5		
Accounting		Operating System	B+	Algorithm Design	3		
Economics		System Analysis & Design	A	Shell Scripting	5		
		Introduction to Computer Organisation	A	Cloud Platform (AWS/Azure/GCP)	5		
		ICT & Islam	A-	Containerisation (Docker/K8s)	7		
		Digital & Embedded Systems	C+				
		Enterprise Networks					
		Intelligent Systems	B+				
		Computer Architecture & Assembly Language	B+				
		Cross-Platform Software Development					
		Computation and Complexity					
		User Experience Design					

Table A.6: Student Profile and Assessment Data of Student 6

A.7 Student 7

GENDER		AGE		CURRENT SEMESTER		CURRENT SPECIALISATION	
Male		21		5		- Networking & Data Communication - Security in Digital System	
SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A	Programming Fundamentals	A-	Python Programming	8	Realistic	22
Additional Mathematics	B+	Data Structure and Algorithm	B+	SQL & Database management	7	Investigative	22
Physics	B+	Database System	B-	Git Version Control	8	Artistic	18
Computer Science		Computer Networking	B-	JavaScript / Web Development	8	Social	19
English	A+	Object Oriented Programming	B+	Linux / Command Line	6	Enterprising	13
Chemistry	A-	Discrete Structures		Problem Solving Skills	7	Conventional	21
Biology	A	Software Engineering	A	Debugging Skills	6		
Accounting		Operating System	B+	Algorithm Design	6		
Economics		System Analysis & Design	B+	Shell Scripting	4		

	Introduction to Computer Organisation	A	Cloud Platform (AWS/Azure/GCP)	3	
	ICT & Islam		Containerisation (Docker/K8s)	2	
	Digital & Embedded Systems	A			
	Enterprise Networks				
	Intelligent Systems	A			
	Computer Architecture & Assembly Language				
	Cross-Platform Software Development				
	Computation and Complexity				
	User Experience Design				

Table A.7: Student Profile and Assessment Data of Student 7

A.8 Student 8

GENDER	AGE	CURRENT SEMESTER	CURRENT SPECIALISATION
Male	21	5	- Networking & Data Communication - Security in Digital System

SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A+	Programming Fundamentals	B+	Python Programming	6	Realistic	20
Additional Mathematics	B+	Data Structure and Algorithm	A	SQL & Database management	5	Investigative	19
Physics	A-	Database System	B	Git Version Control	5	Artistic	18
Computer Science		Computer Networking	A	JavaScript / Web Development	7	Social	19
English	A-	Object Oriented Programming	B+	Linux / Command Line	6	Enterprising	18
Chemistry	B+	Discrete Structures		Problem Solving Skills	8	Conventional	20
Biology	B+	Software Engineering	B+	Debugging Skills	8		
Accounting		Operating System	B+	Algorithm Design	5		
Economics		System Analysis & Design	A-	Shell Scripting	5		
		Introduction to Computer Organisation	A-	Cloud Platform (AWS/Azure/GCP)	4		
		ICT & Islam		Containerisation (Docker/K8s)	3		
		Digital & Embedded Systems					
		Enterprise Networks	B+				
		Intelligent Systems	A-				

	Computer Architecture & Assembly Language	A-		
	Cross-Platform Software Development			
	Computation and Complexity			
	User Experience Design			

Table A.8: Student Profile and Assessment Data of Student 8

A.9 Student 9

GENDER		AGE		CURRENT SEMESTER		CURRENT SPECIALISATION	
Male		22		5		- Artificial Intelligence - Data Engineering	
SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A+	Programming Fundamentals	A	Python Programming	7	Realistic	18
Additional Mathematics	A	Data Structure and Algorithm	A	SQL & Database management	4	Investigative	20
Physics	A	Database System		Git Version Control	1	Artistic	15
Computer Science		Computer Networking	B+	JavaScript / Web Development	7	Social	18

English	A	Object Oriented Programming	A-	Linux / Command Line	3	Enterprising	20
Chemistry	A-	Discrete Structures		Problem Solving Skills	6	Conventional	18
Biology	A-	Software Engineering	A-	Debugging Skills	5		
Accounting		Operating System	A-	Algorithm Design	4		
Economics		System Analysis & Design	A	Shell Scripting	4		
		Introduction to Computer Organisation	A	Cloud Platform (AWS/Azure/GCP)	4		
		ICT & Islam		Containerisation (Docker/K8s)	4		
		Digital & Embedded Systems					
		Enterprise Networks					
		Intelligent Systems	A				
		Computer Architecture & Assembly Language					
		Cross-Platform Software Development					
		Computation and Complexity					
		User Experience Design					

Table A.9: Student Profile and Assessment Data of Student 9

A.10 Student 10

GENDER		AGE		CURRENT SEMESTER		CURRENT SPECIALISATION	
Male		22		5		- Networking & Data Communication - Security in Digital System	
SPM RESULTS		PREVIOUS SEMESTER RESULT		TECHNICAL SKILLS		RIASEC PERSONALITY TRAITS	
Modern Mathematics	A-	Programming Fundamentals	A	Python Programming	7	Realistic	22
Additional Mathematics	C	Data Structure and Algorithm	B+	SQL & Database management	6	Investigative	22
Physics	D	Database System	B+	Git Version Control	6	Artistic	21
Computer Science	A	Computer Networking	A	JavaScript / Web Development	6	Social	19
English	A-	Object Oriented Programming	C+	Linux / Command Line	6	Enterprising	24
Chemistry	D	Discrete Structures	A	Problem Solving Skills	8	Conventional	23
Biology	D	Software Engineering	B+	Debugging Skills	7		
Accounting	A	Operating System	B+	Algorithm Design	7		
Economics	A	System Analysis & Design	A	Shell Scripting	7		
		Introduction to Computer Organisation	A	Cloud Platform (AWS/Azure/GCP)	7		

	ICT & Islam	A	Containerisation (Docker/K8s)	6
	Digital & Embedded Systems	A		
	Enterprise Networks	A-		
	Intelligent Systems	B+		
	Computer Architecture & Assembly Language	B+		
	Cross-Platform Software Development			
	Computation and Complexity			
	User Experience Design			

Table A.10: Student Profile and Assessment Data of Student 10

B. GANTT CHART

PHASE	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13
DATE	06-Oct	13-Oct	20-Oct	27-Oct	03-Nov	10-Nov	17-Nov	01-Dec	08-Dec	15-Dec	22-Dec	29-Dec	05-Jan
Topic Selection													
Literature Review													
Model Design													
Beta System Development													
Data Collection (Part 1)													
Beta System Testing													
Expert Consultation 1													
Data Collection (Part 2)													
Test Updated Beta System													
True System development													
Expert Consultation 2													
Comparison & Analysis													
Drafting Final Report													
Documents Submission													
Final Presentation													

Figure B: Gantt Chart of Project Milestones and Development

C. DIVISION OF WORK.

Name	Work Distribution
Arif Danial	Project Planning & Management, Literature Review, System Development (Frontend & Backend), System Debugging, Report Chapter 1, Report Chapter 4, Report Appendices, Report Finalisation, Poster Design & Creation, Logbook
Syazwan fariz	Literature Review, Survey Questions Creation, Data Collection, Assist In System Development, Report Chapter 2, Report Chapter 3, Technical Report Writing

D. SOURCE CODE RESPIRATORY

The full deployment code (Flask web app) is available on GitHub.

Github respiratory: <https://github.com/XxNergigantexX/FYP1.git>

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